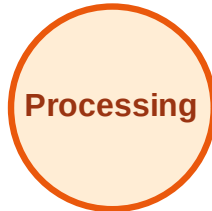


BFS Traversal

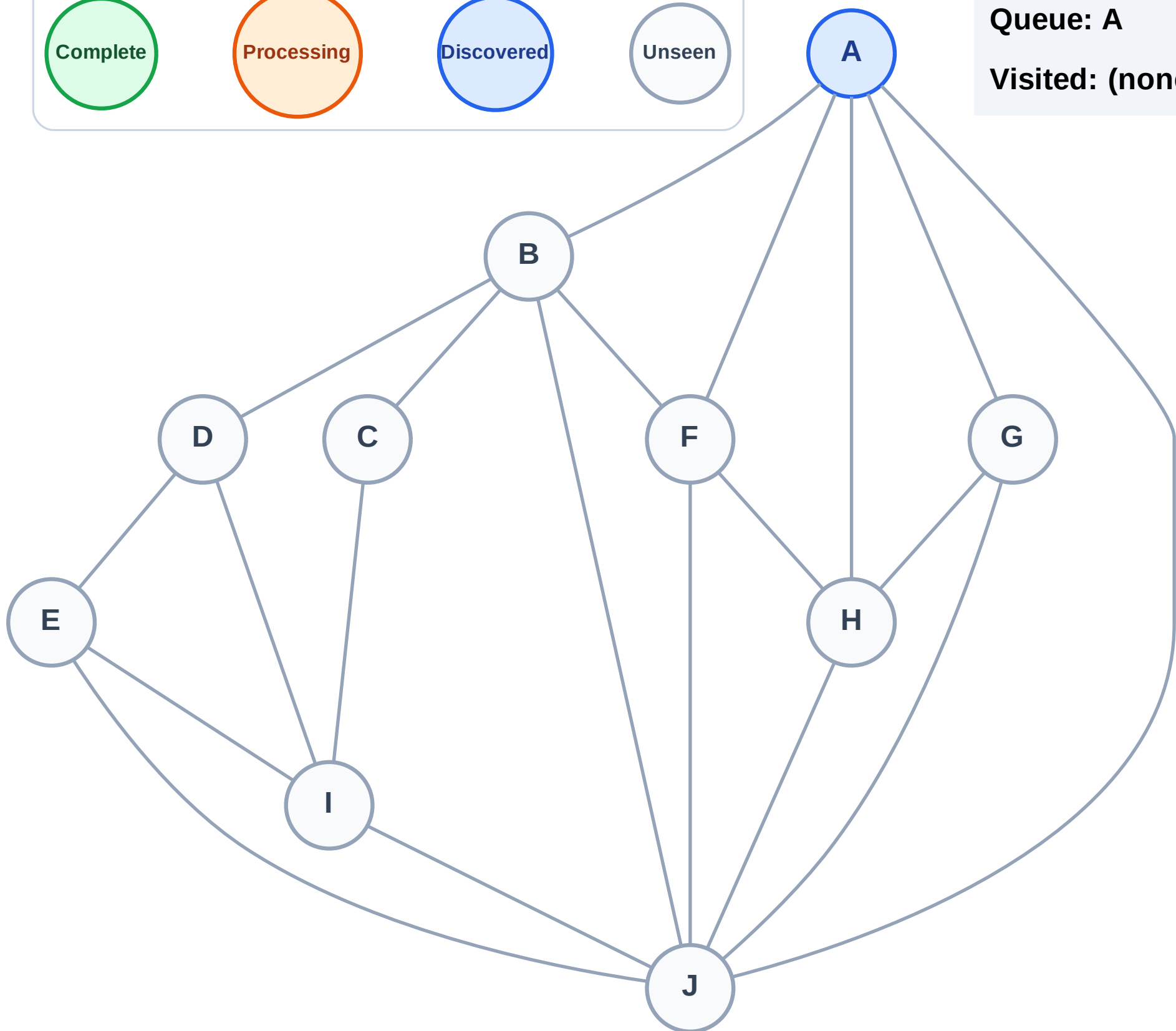
Step 1: Start at A

Legend



Queue: A

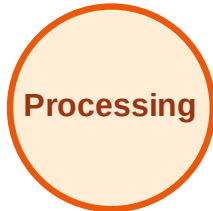
Visited: (none)



BFS Traversal

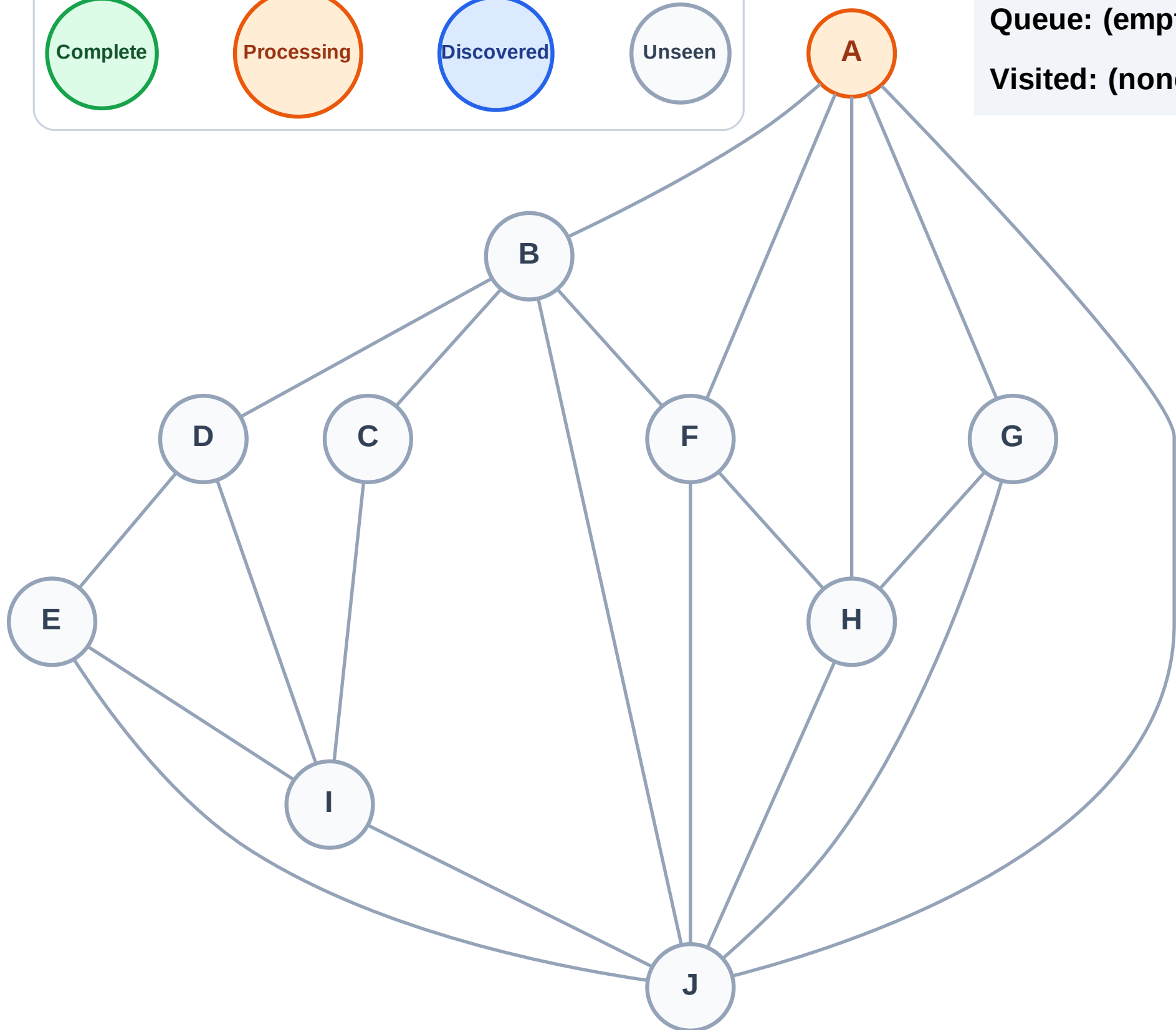
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, F, G, H, J from A

Legend

Complete

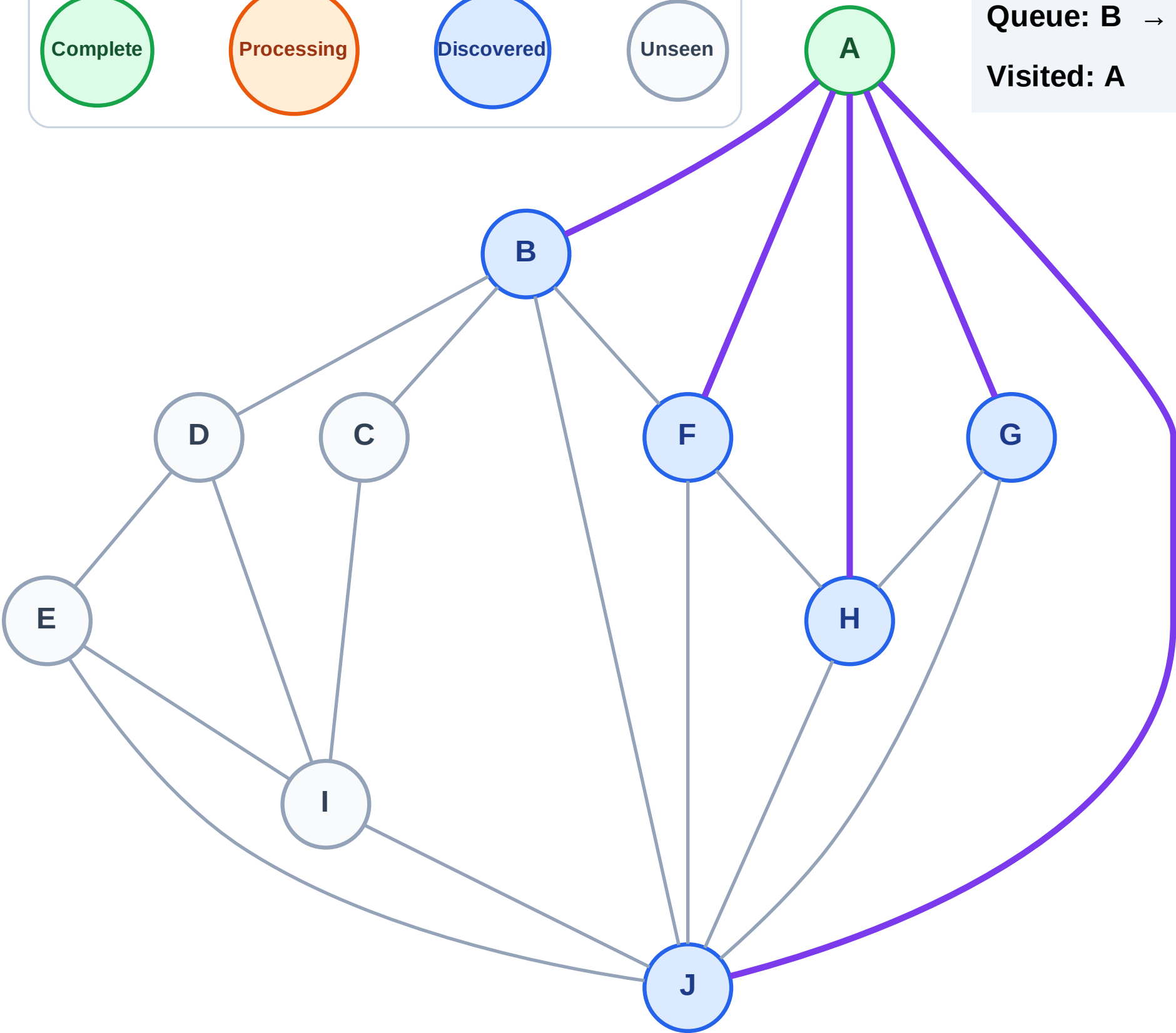
Processing

Discovered

Unseen

Queue: B → F → G → H → J

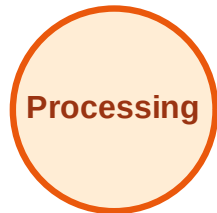
Visited: A



BFS Traversal

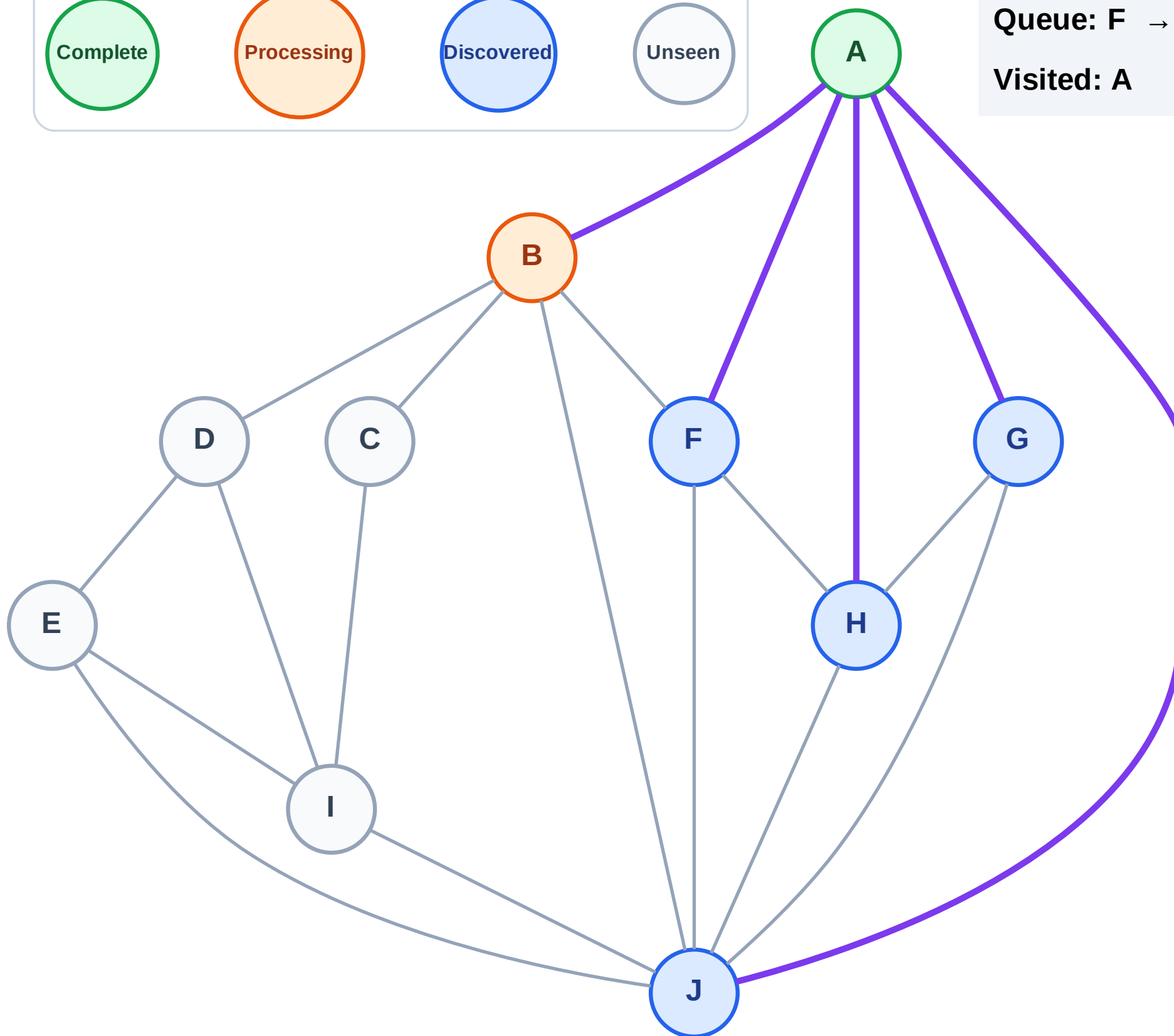
Step 4: Process node B

Legend



Queue: F → G → H → J

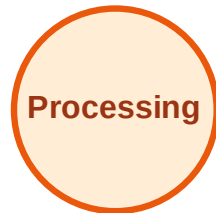
Visited: A



BFS Traversal

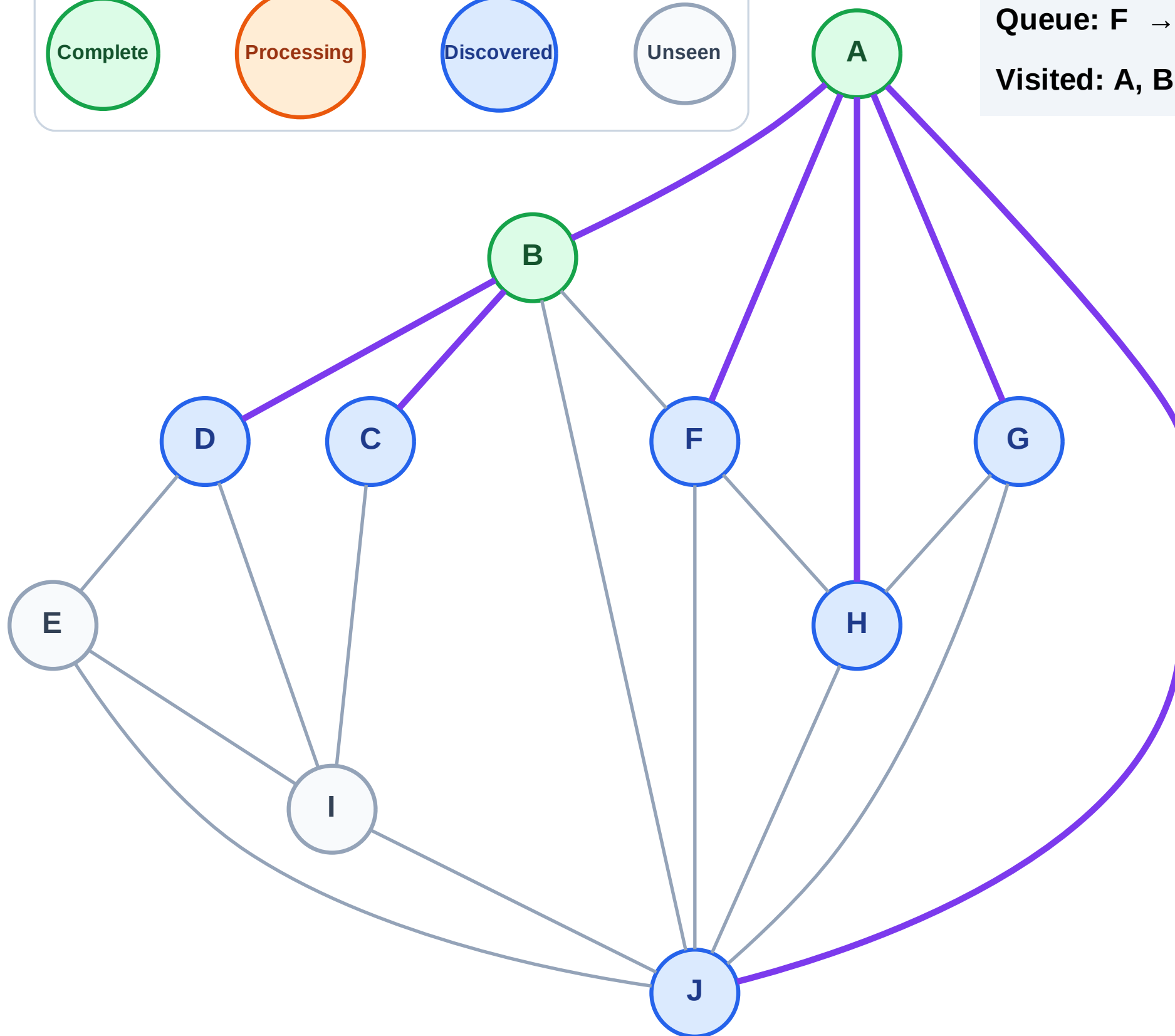
Step 5: Discover C, D from B

Legend



Queue: F → G → H → J → C → D

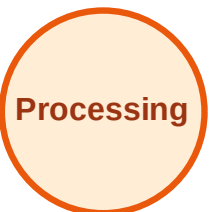
Visited: A, B



BFS Traversal

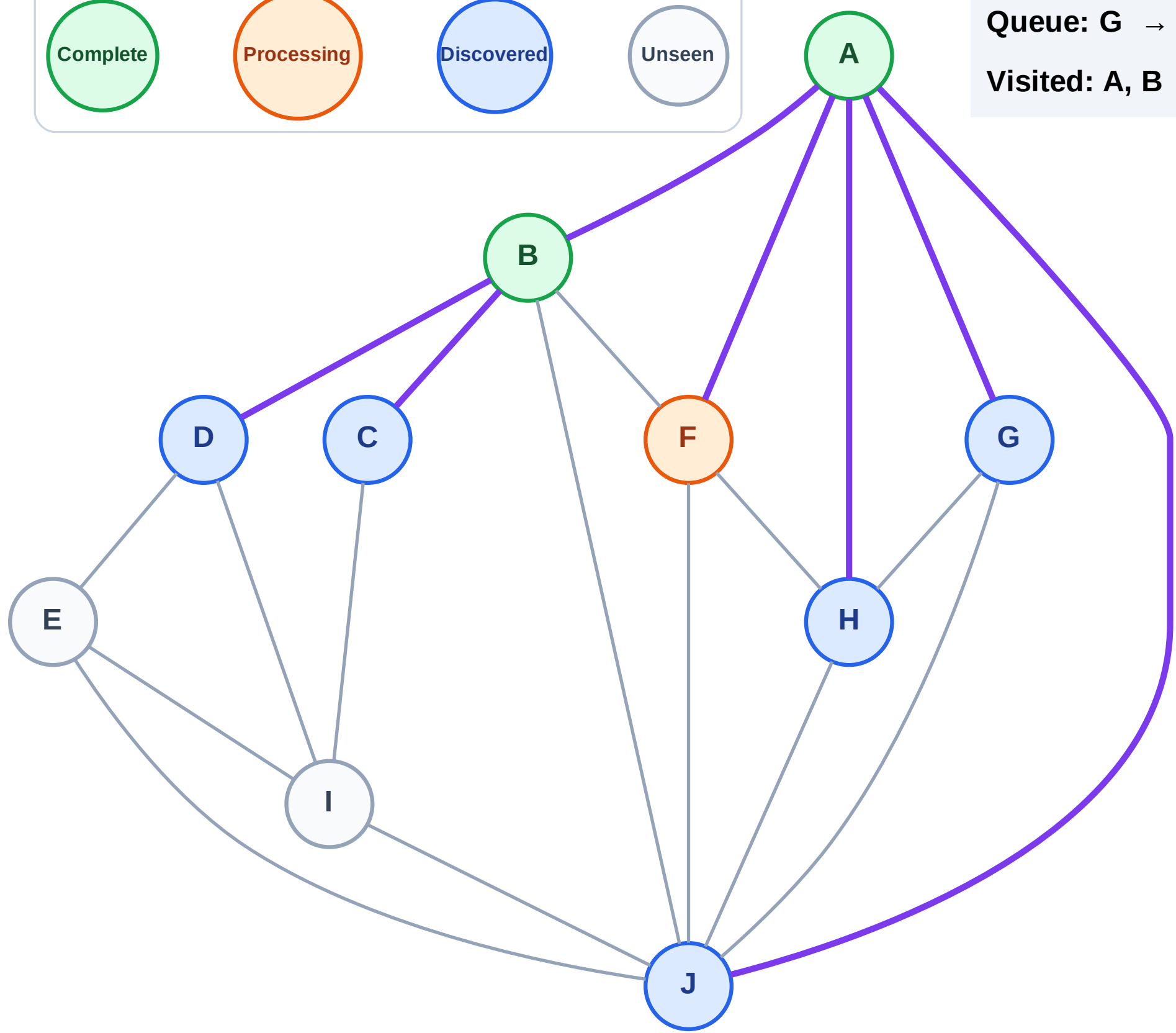
Step 6: Process node F

Legend



Queue: G → H → J → C → D

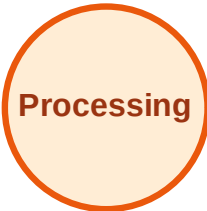
Visited: A, B



BFS Traversal

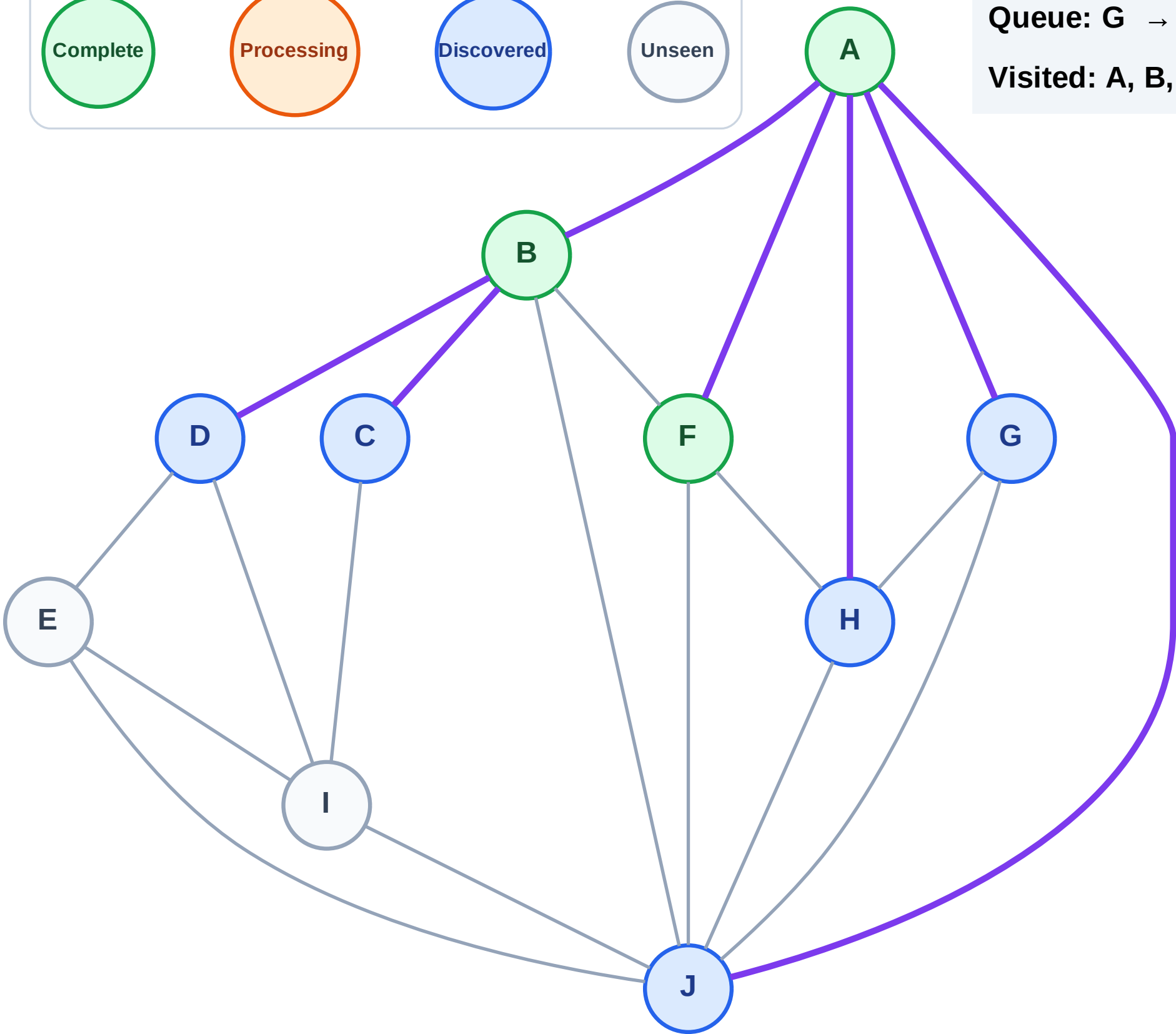
Step 7: No new neighbors from F

Legend



Queue: G → H → J → C → D

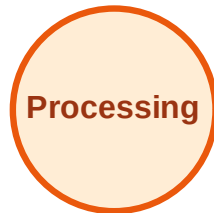
Visited: A, B, F



BFS Traversal

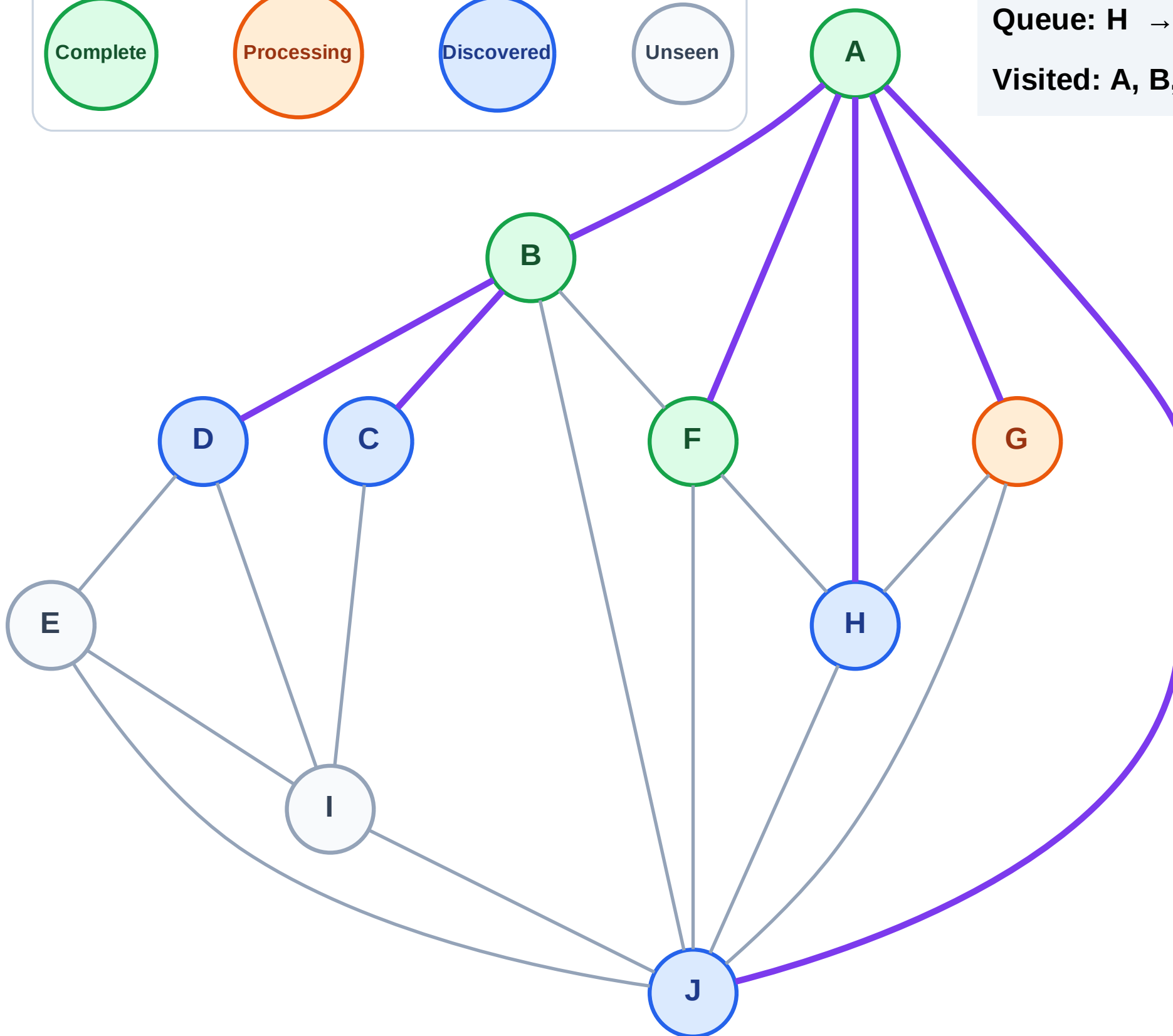
Step 8: Process node G

Legend



Queue: H → J → C → D

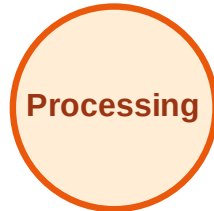
Visited: A, B, F



BFS Traversal

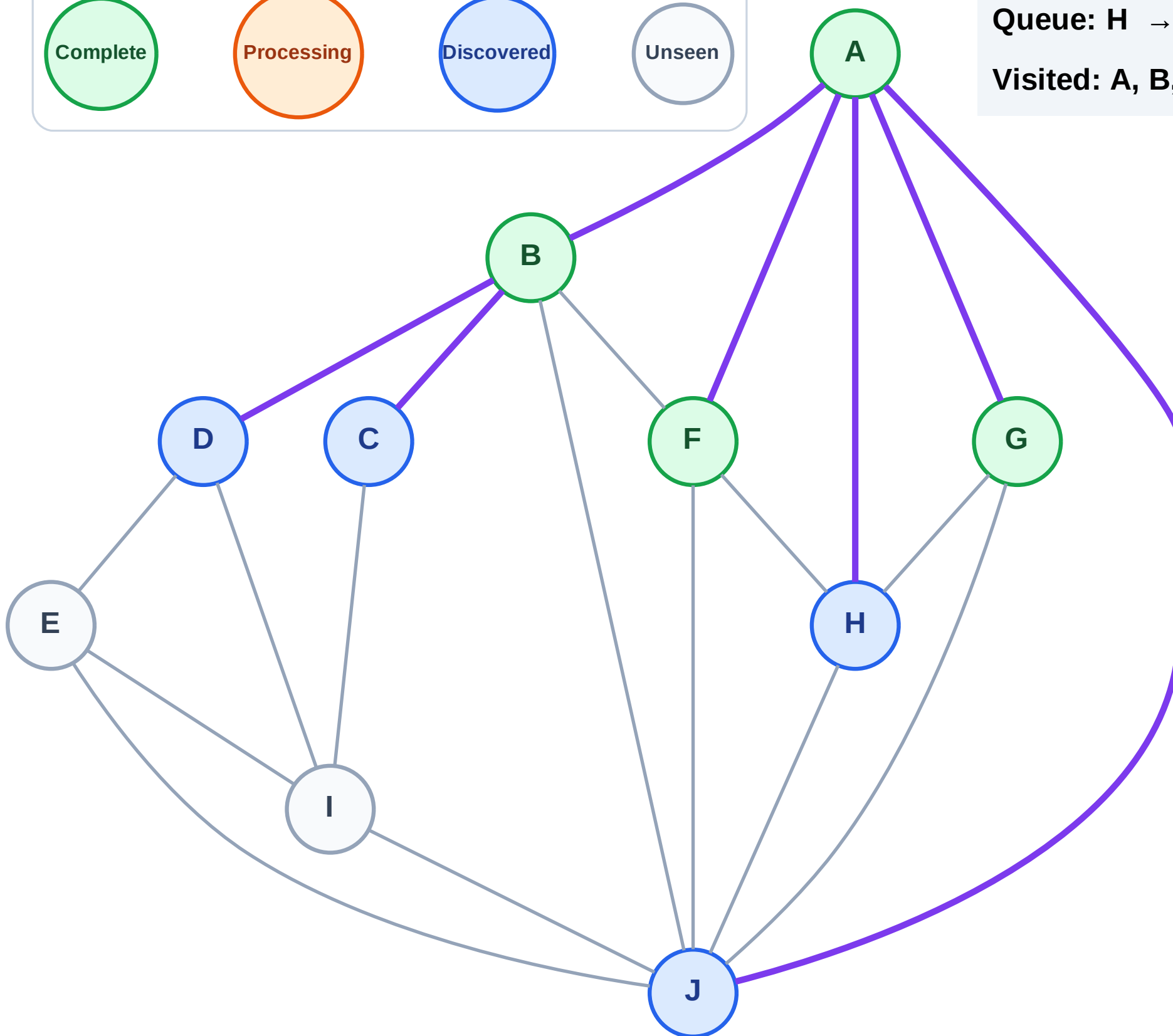
Step 9: No new neighbors from G

Legend



Queue: H → J → C → D

Visited: A, B, F, G



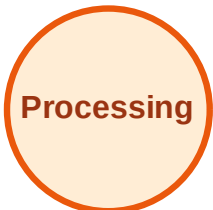
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



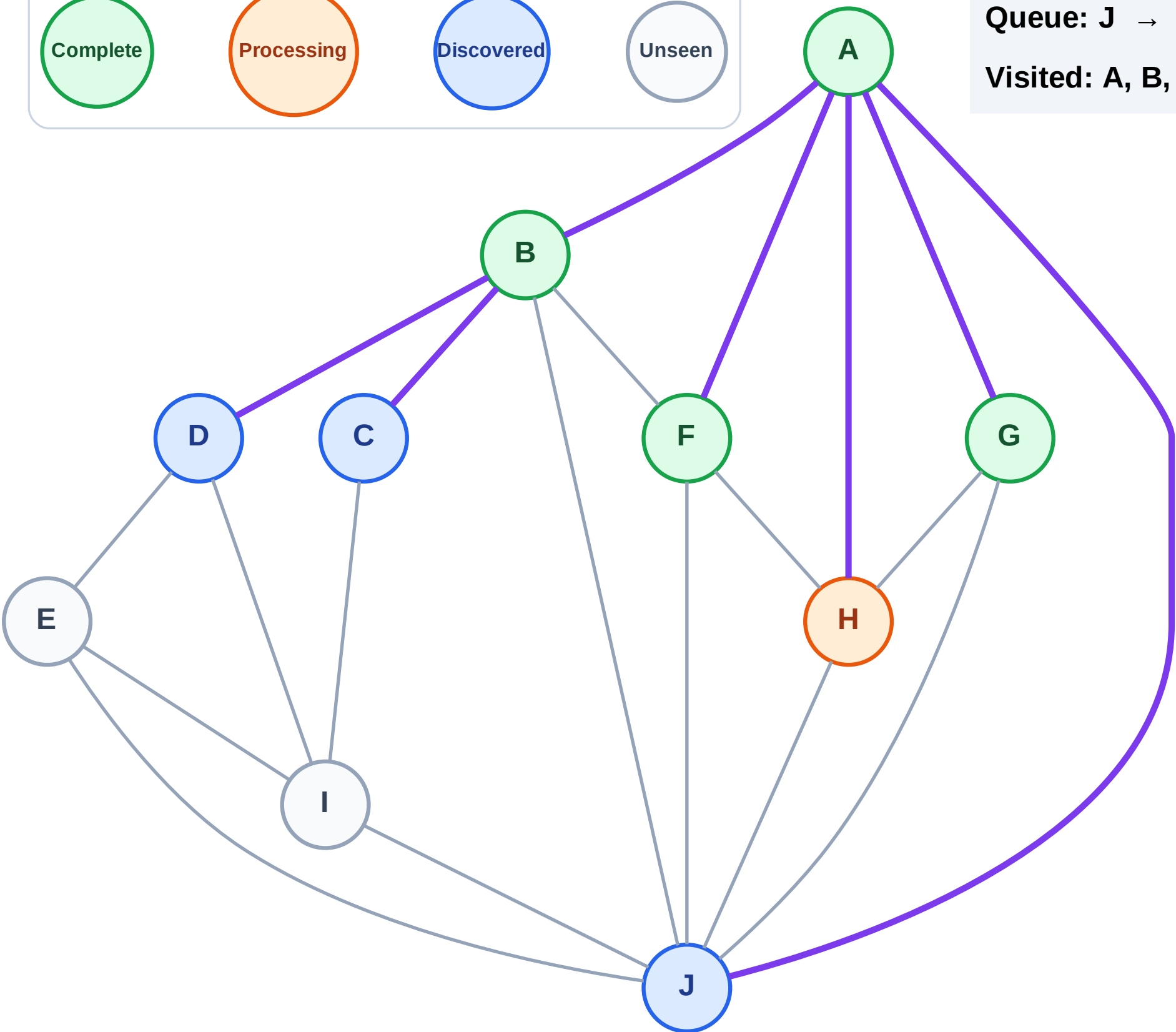
Discovered



Unseen

Queue: J → C → D

Visited: A, B, F, G



BFS Traversal

Step 11: No new neighbors from H

Legend

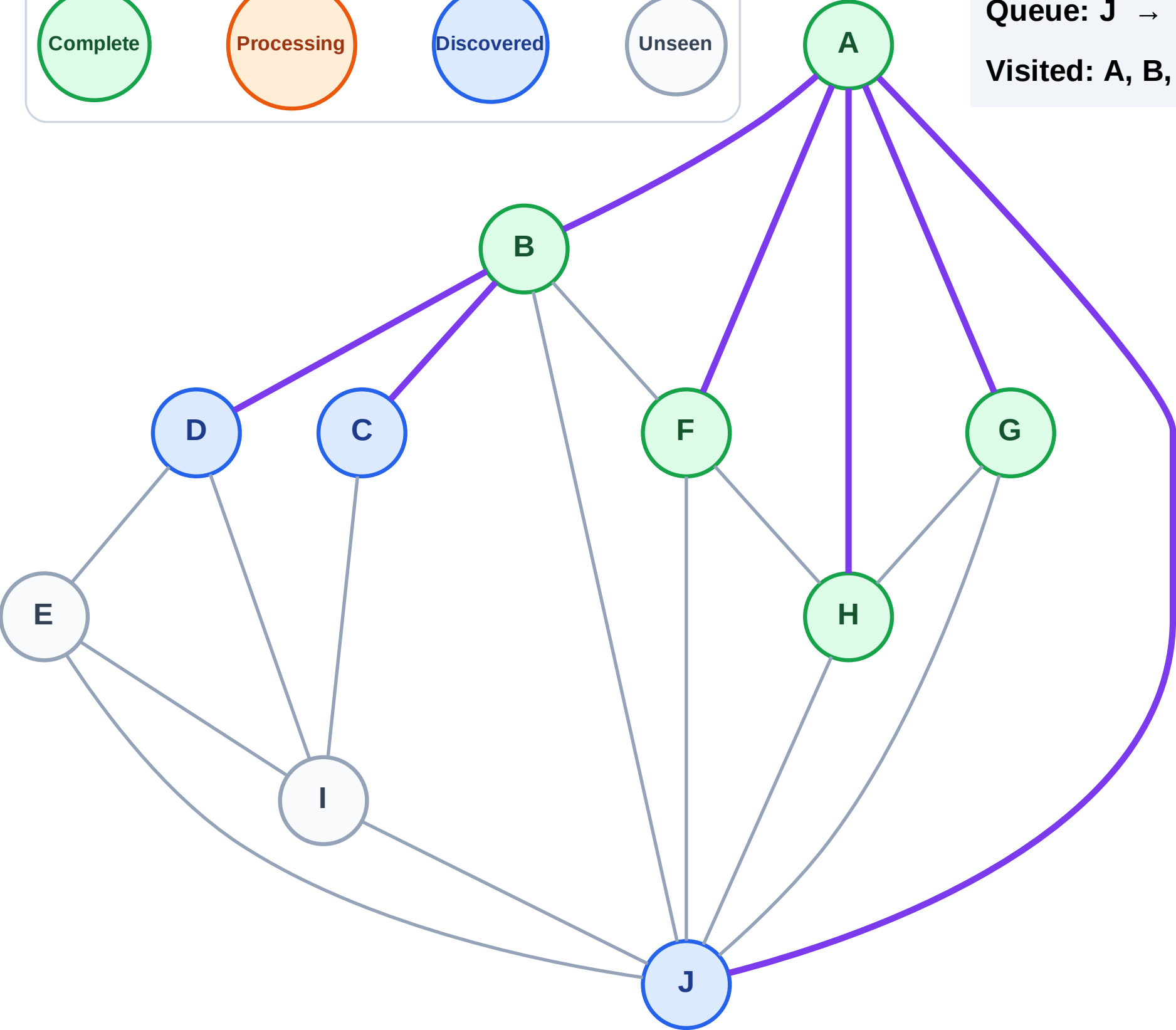
Complete

Processing

Discovered

Unseen

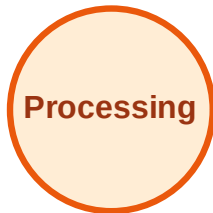
Queue: J → C → D
Visited: A, B, F, G, H



BFS Traversal

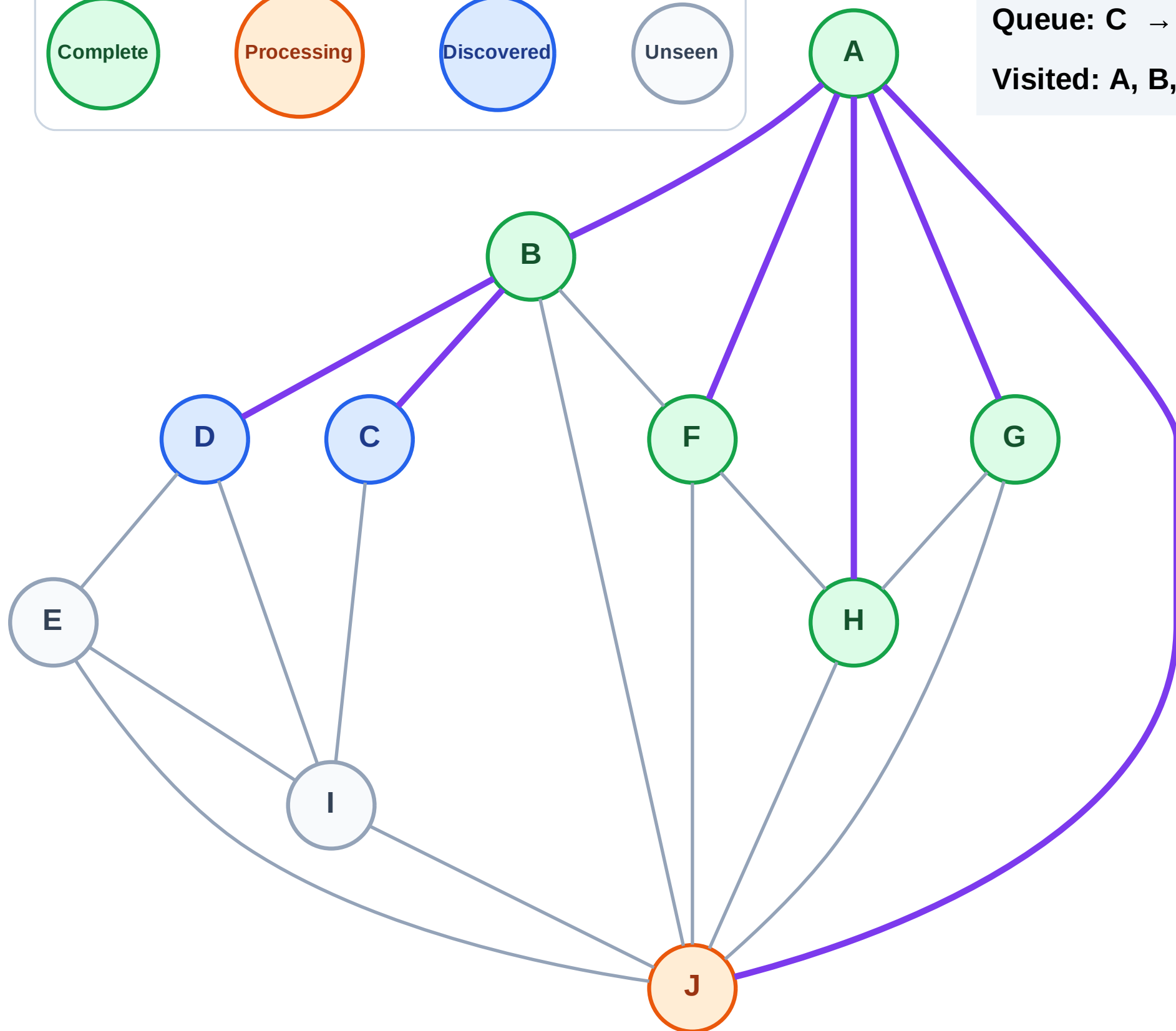
Step 12: Process node J

Legend



Queue: C → D

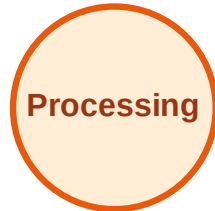
Visited: A, B, F, G, H



BFS Traversal

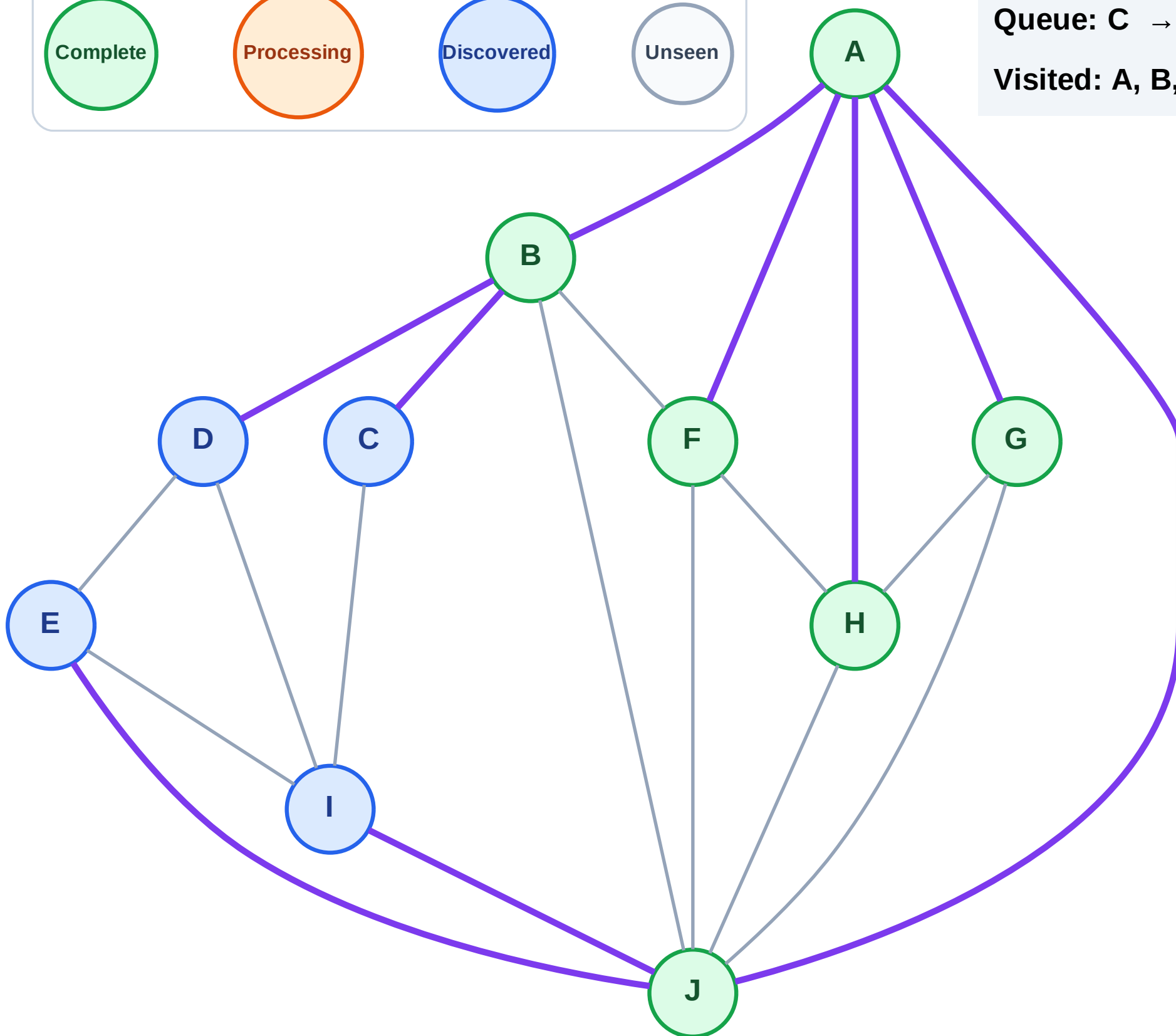
Step 13: Discover E, I from J

Legend



Queue: C → D → E → I

Visited: A, B, F, G, H, J



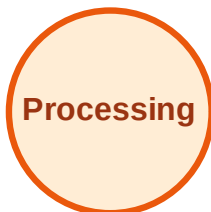
BFS Traversal

Step 14: Process node C

Legend



Complete



Processing



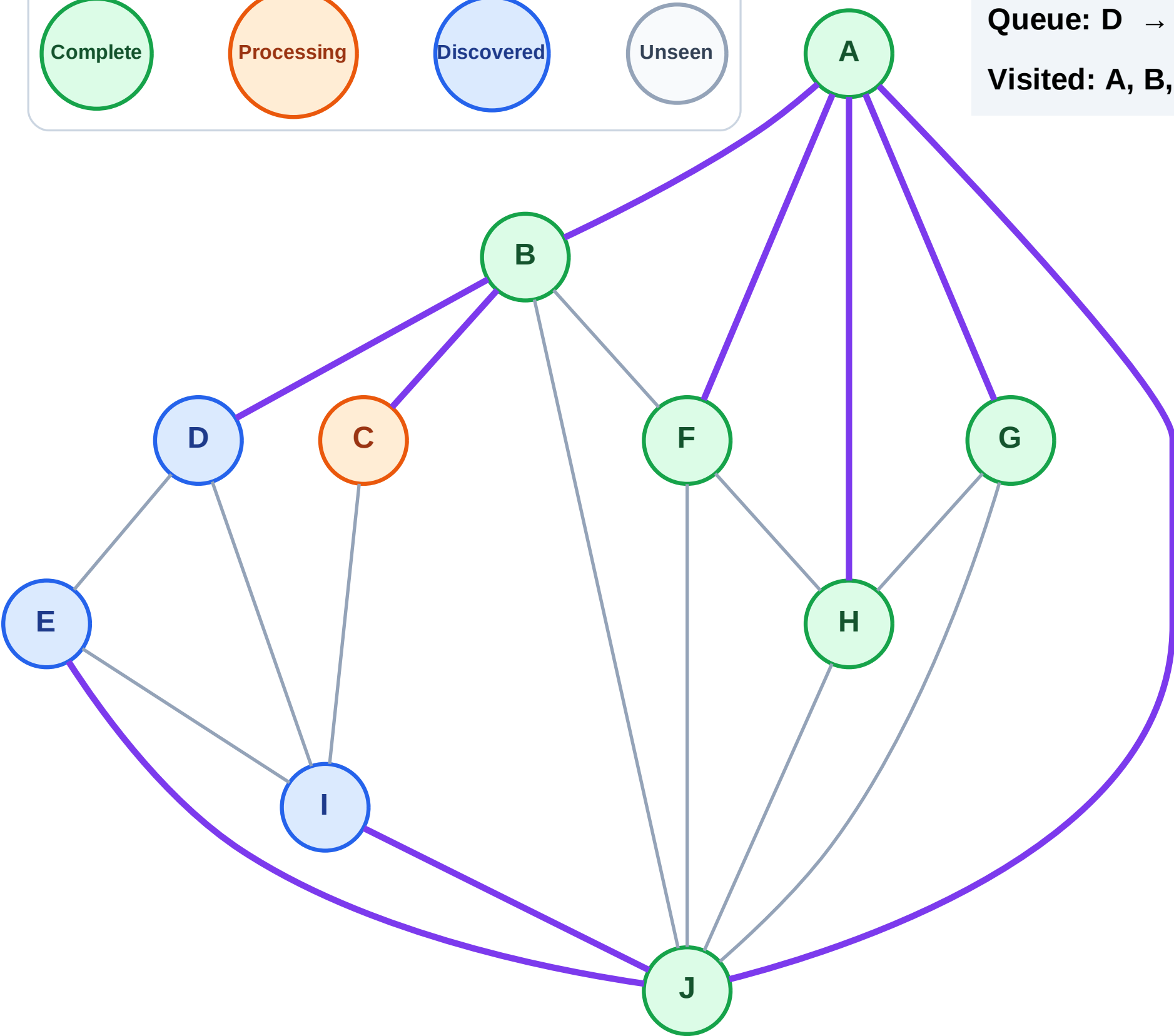
Discovered



Unseen

Queue: D → E → I

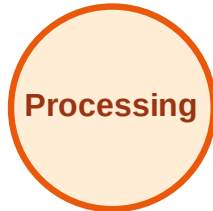
Visited: A, B, F, G, H, J



BFS Traversal

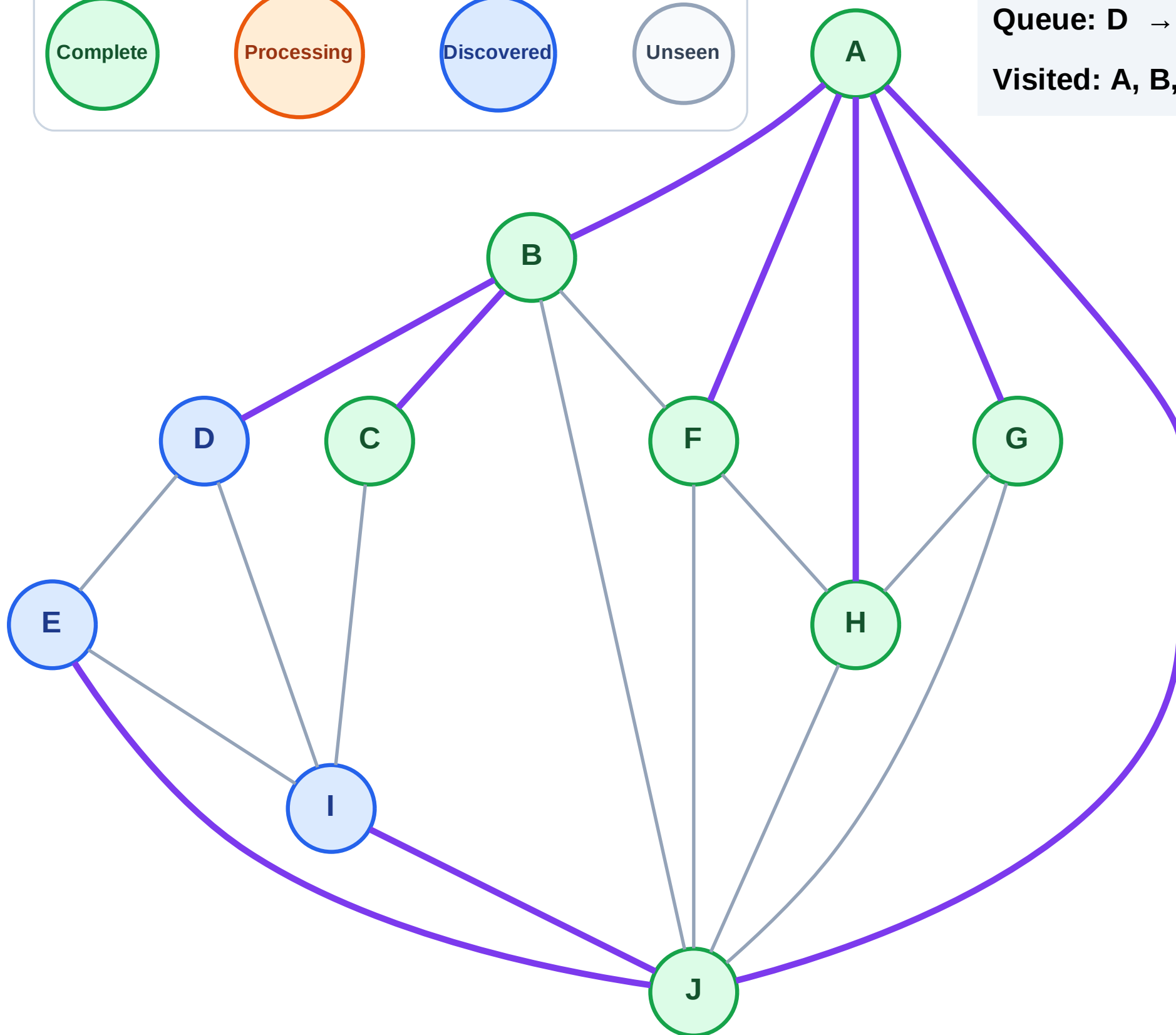
Step 15: No new neighbors from C

Legend



Queue: D → E → I

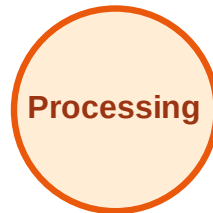
Visited: A, B, C, F, G, H, J



BFS Traversal

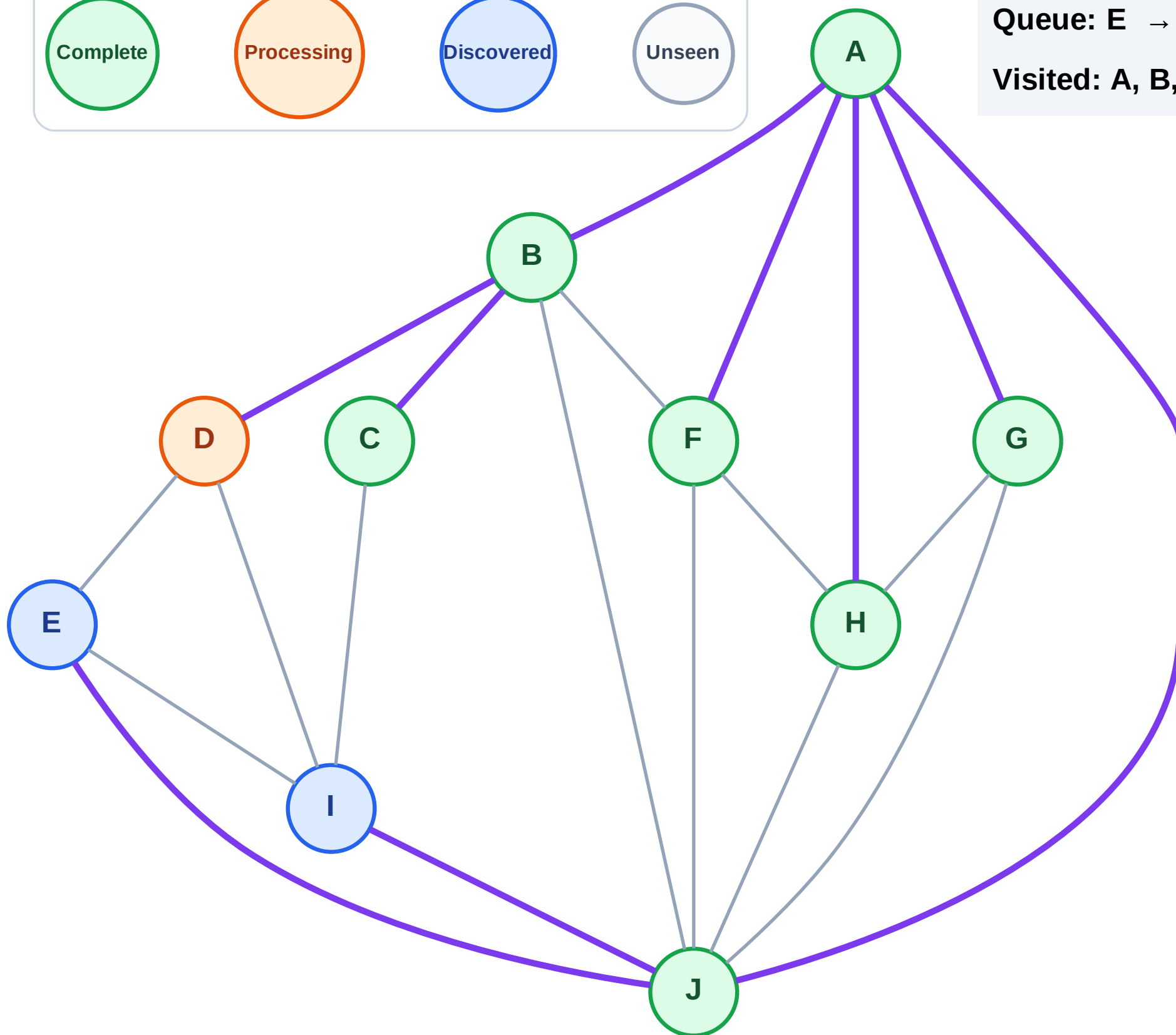
Step 16: Process node D

Legend



Queue: E → I

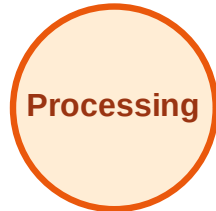
Visited: A, B, C, F, G, H, J



BFS Traversal

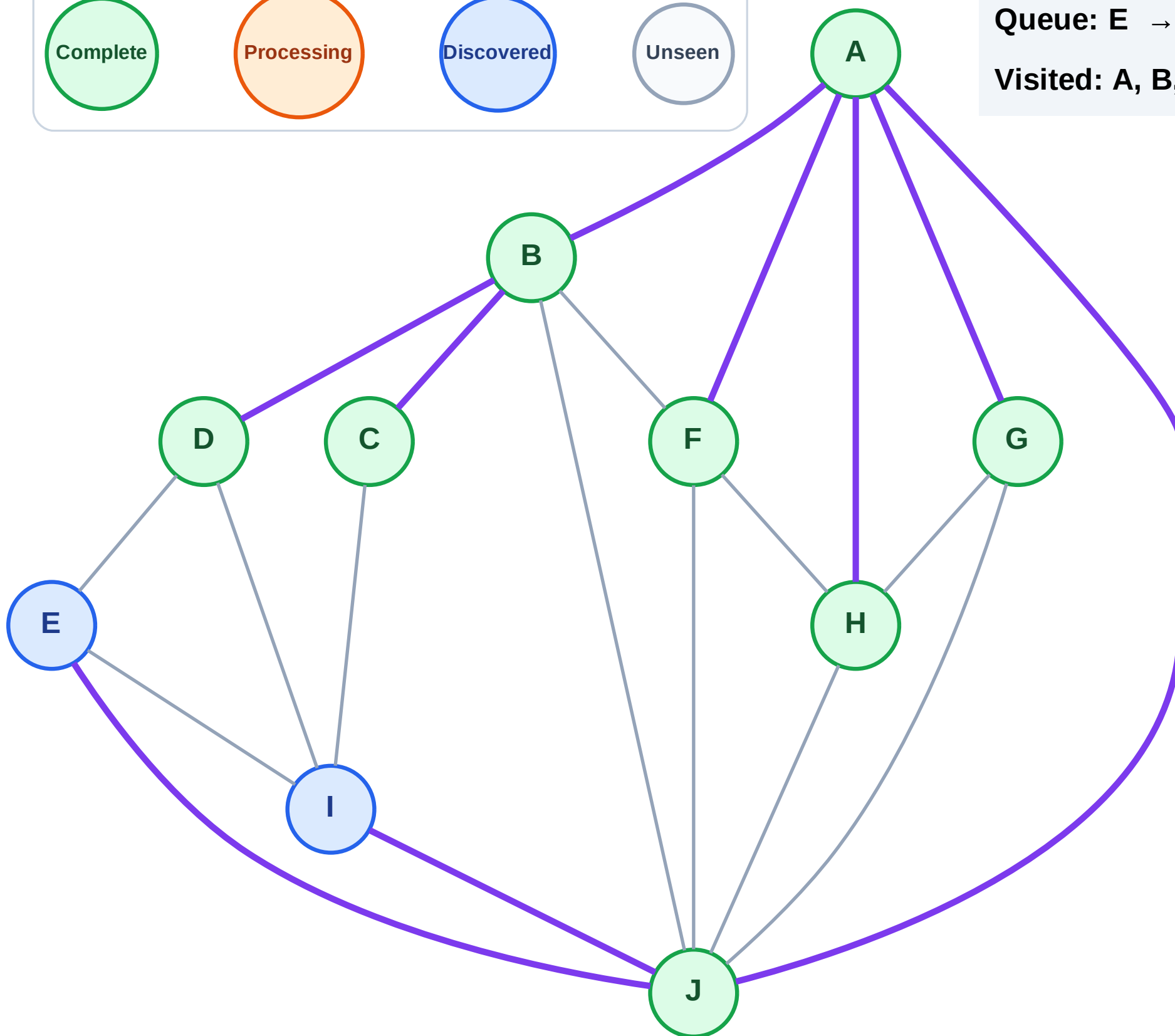
Step 17: No new neighbors from D

Legend



Queue: E → I

Visited: A, B, C, D, F, G, H, J



BFS Traversal

Step 18: Process node E

Legend

Complete

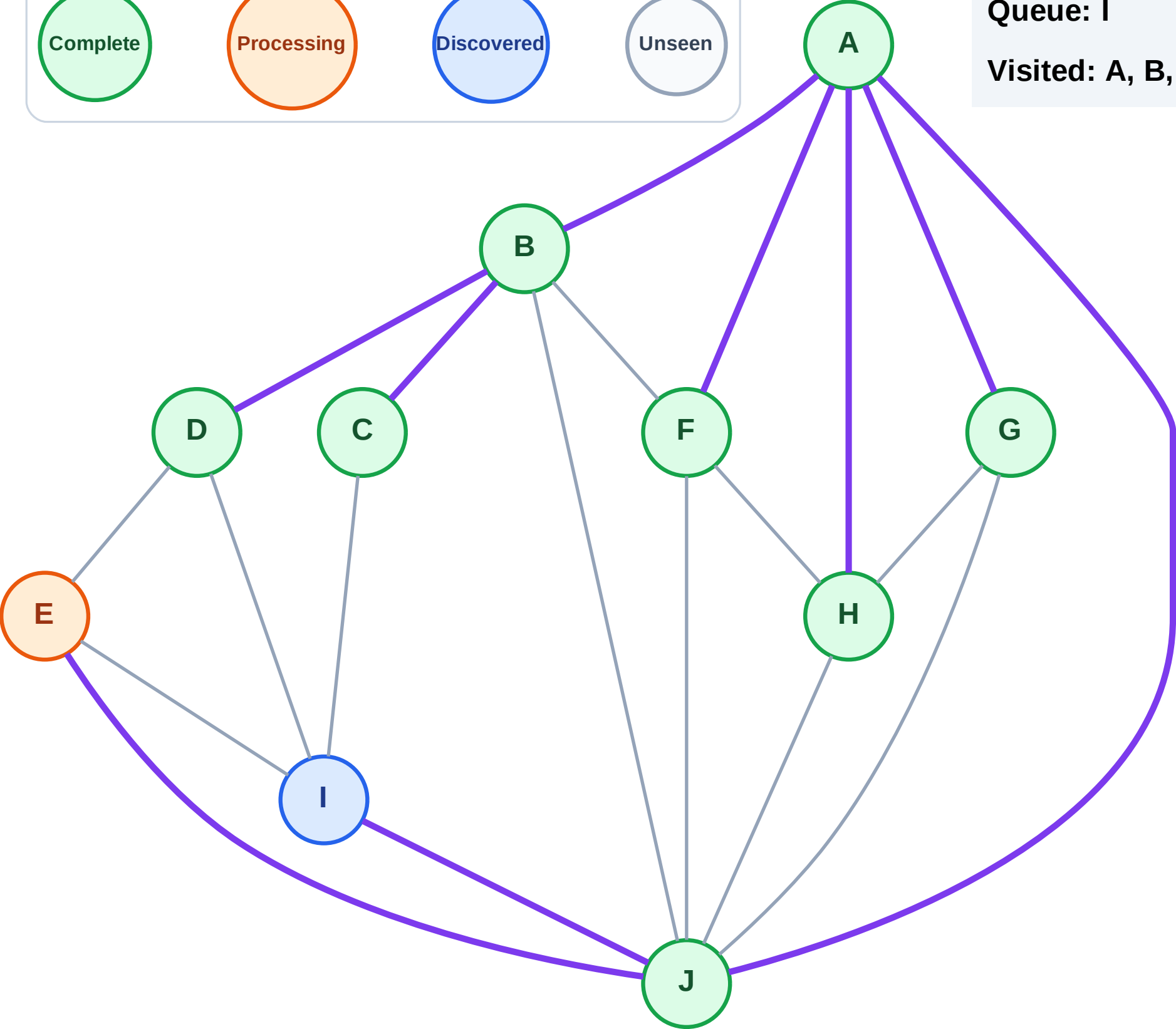
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, F, G, H, J

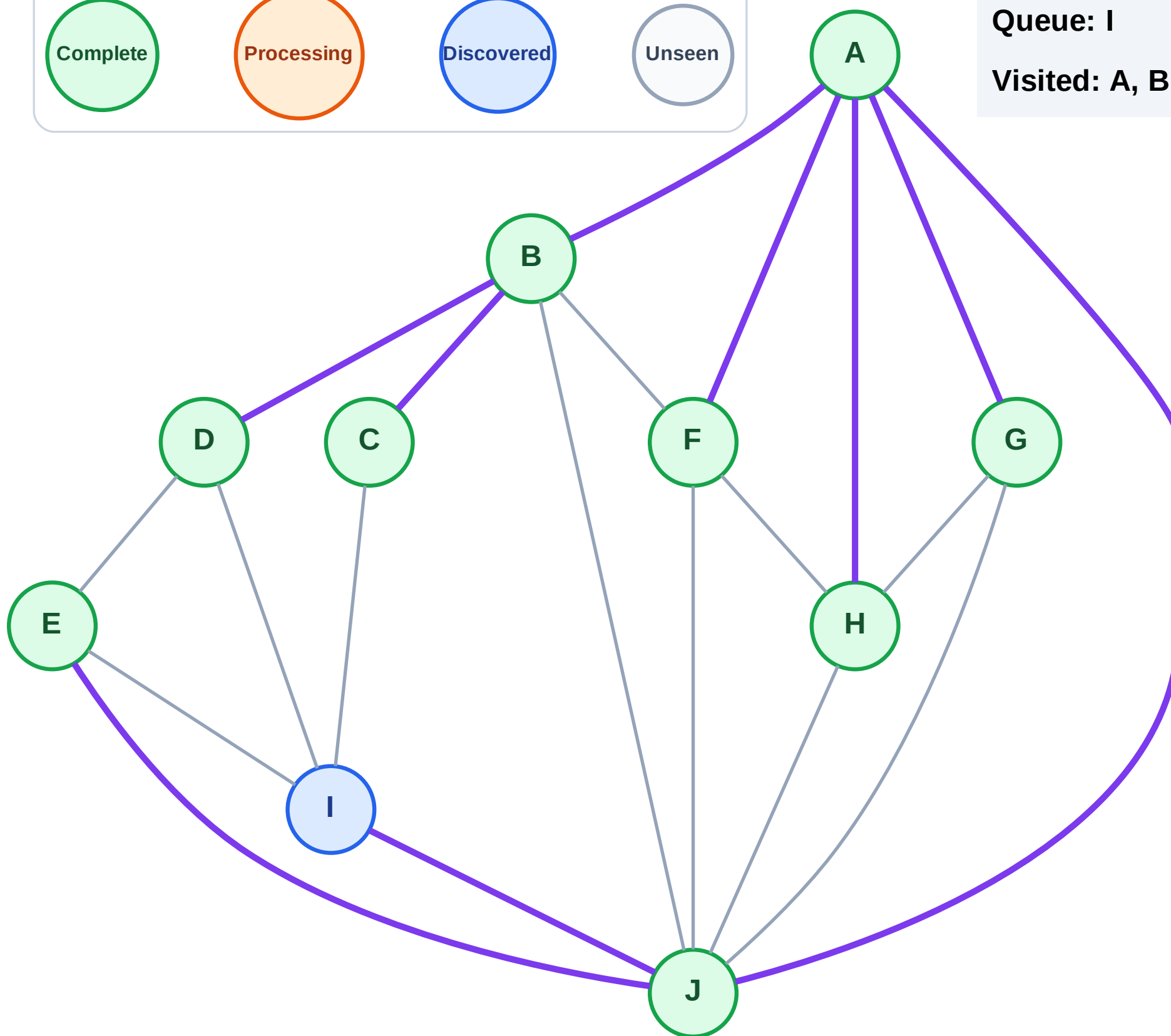


Step 19: No new neighbors from E

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

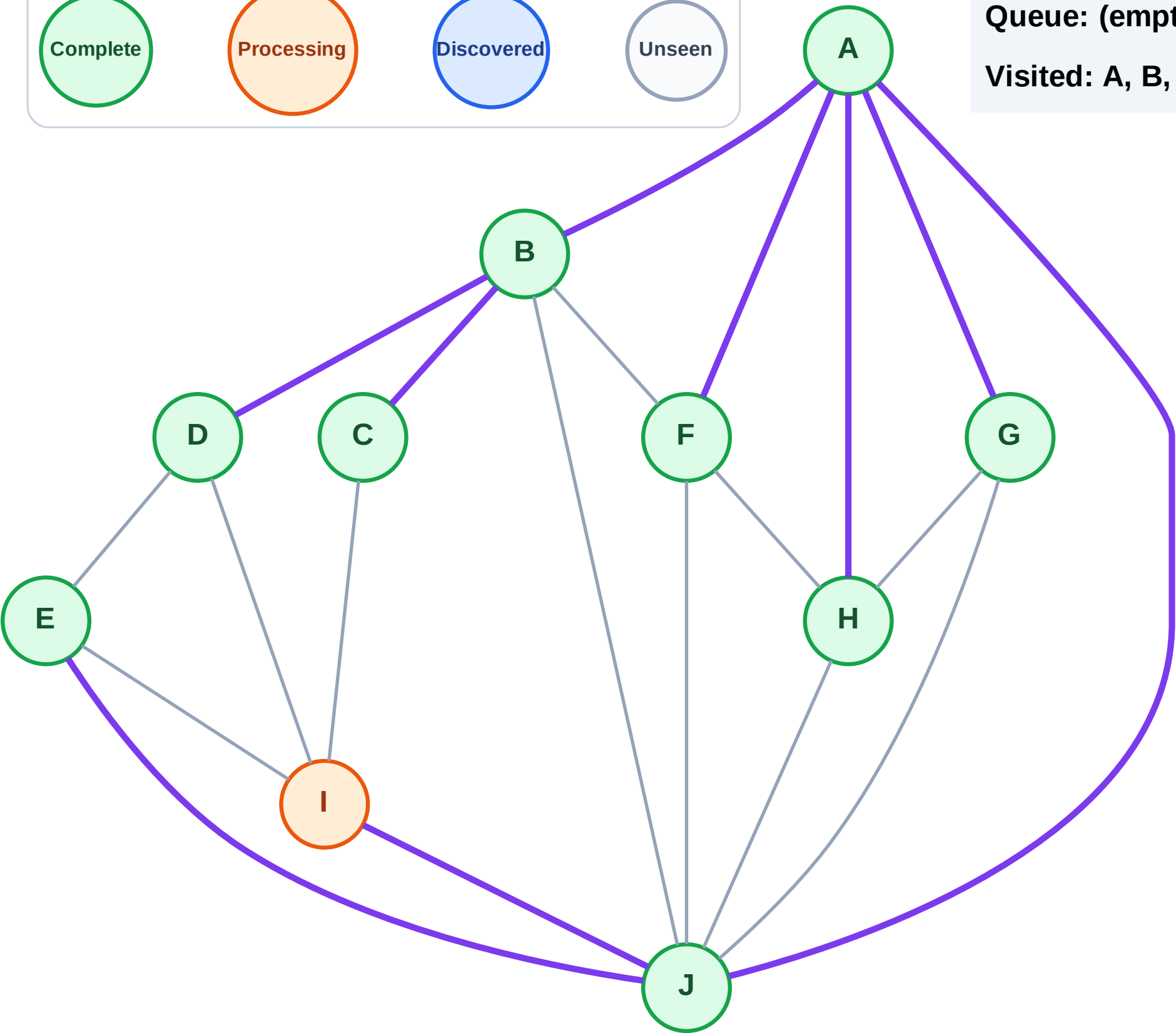
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, J



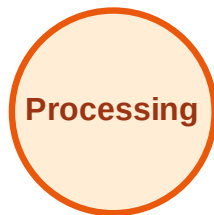
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



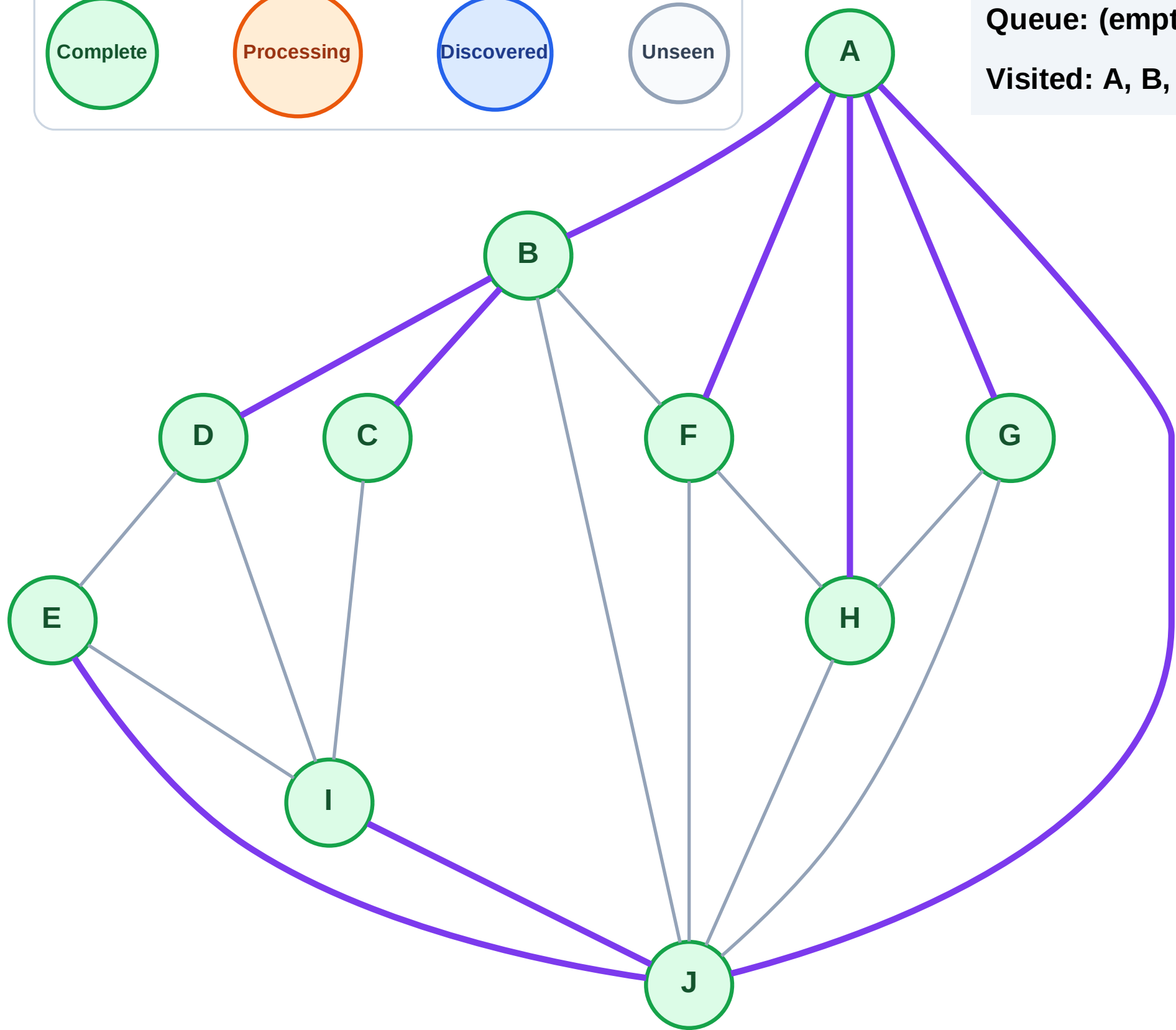
Discovered



Unseen

Queue: (empty)

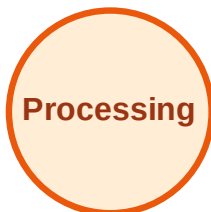
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

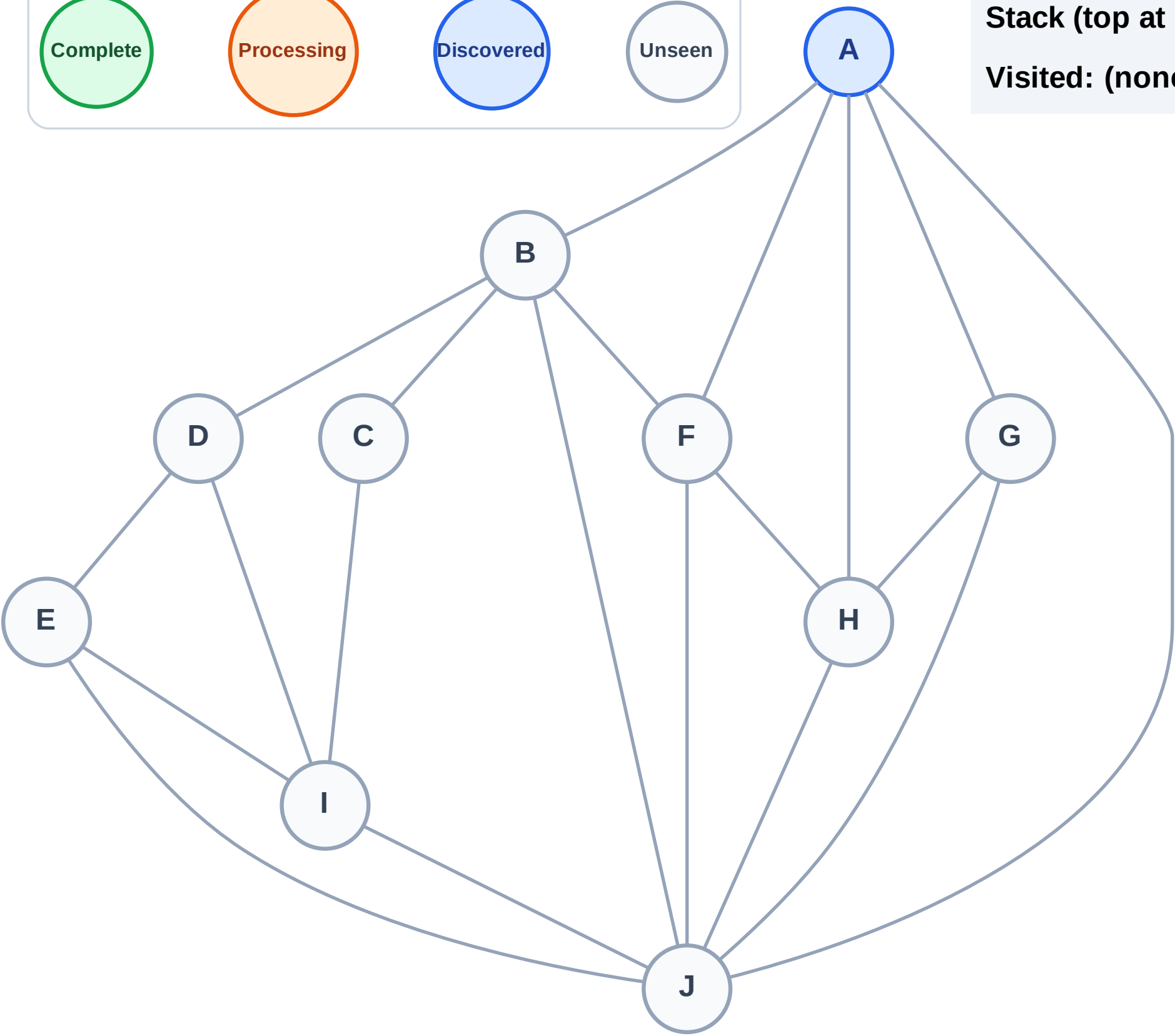
Step 1: Start at A

Legend



Stack (top at right): A

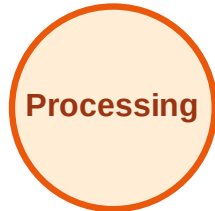
Visited: (none)



DFS Traversal

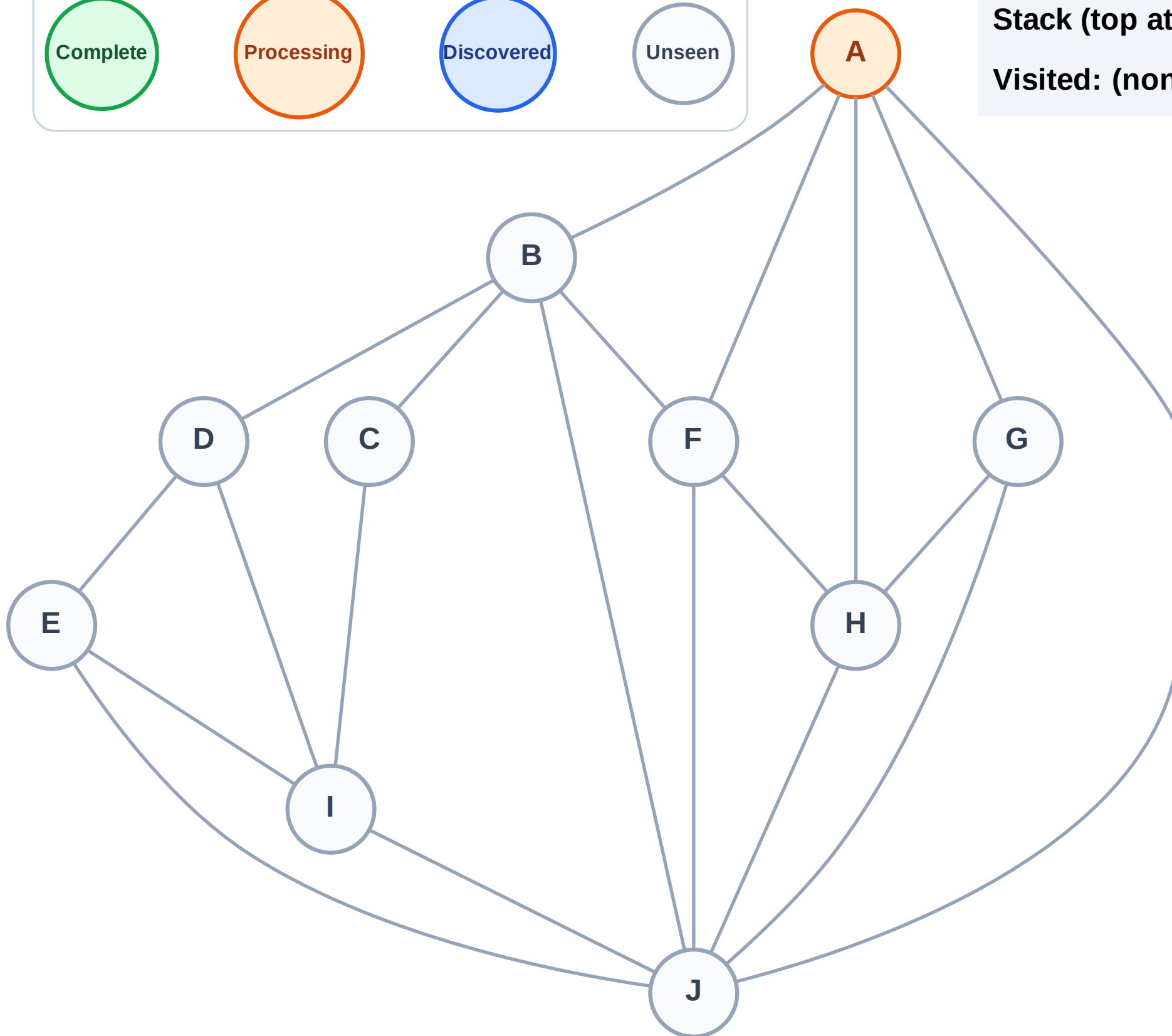
Step 2: Process node A

Legend



Stack (top at right): (empty)

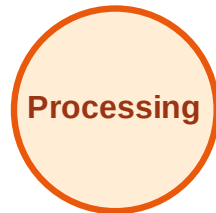
Visited: (none)



DFS Traversal

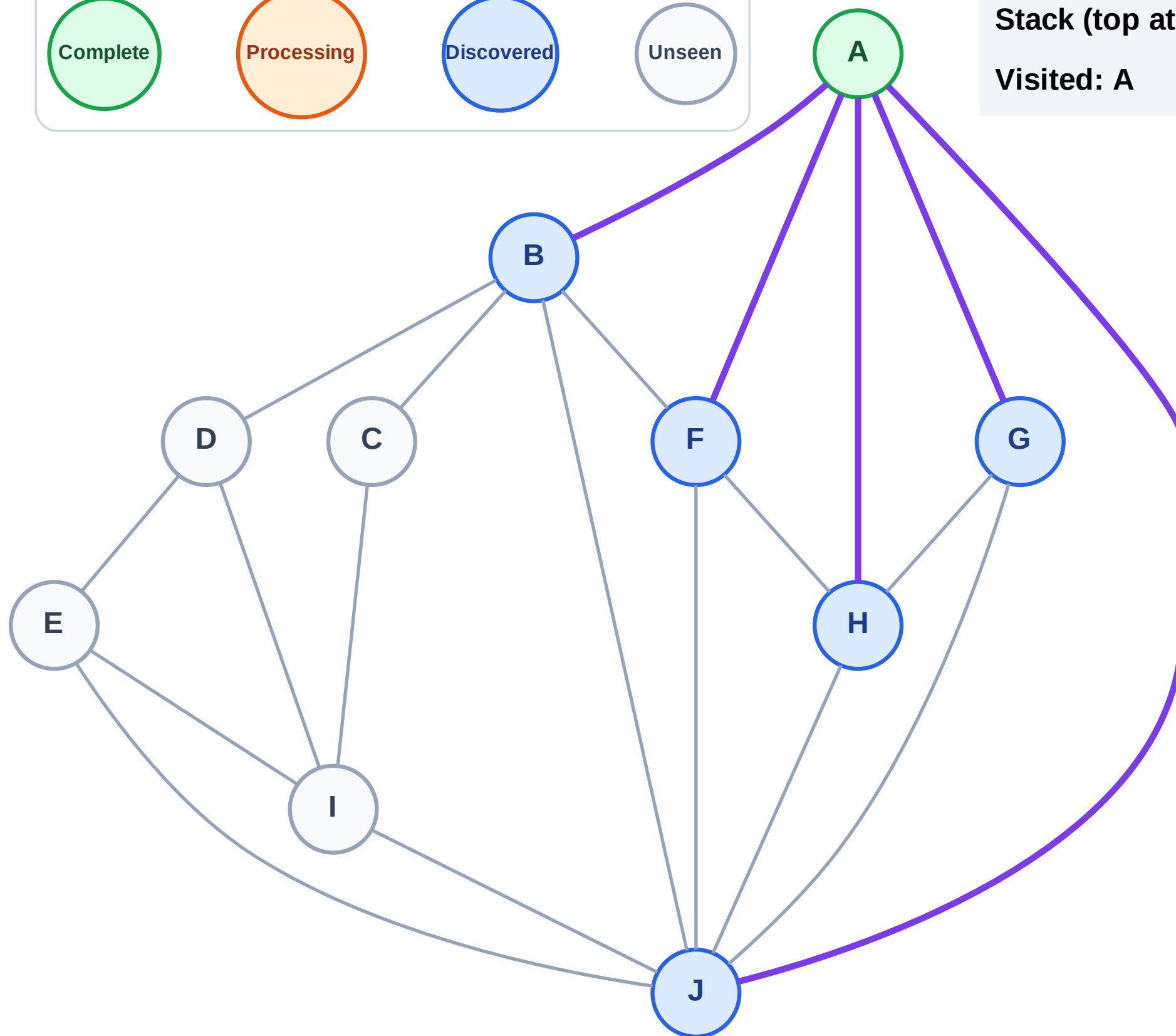
Step 3: Discover J, H, G, F, B from A

Legend



Stack (top at right): J | H | G | F | B

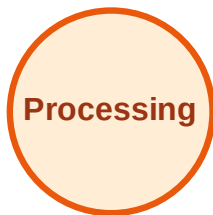
Visited: A



DFS Traversal

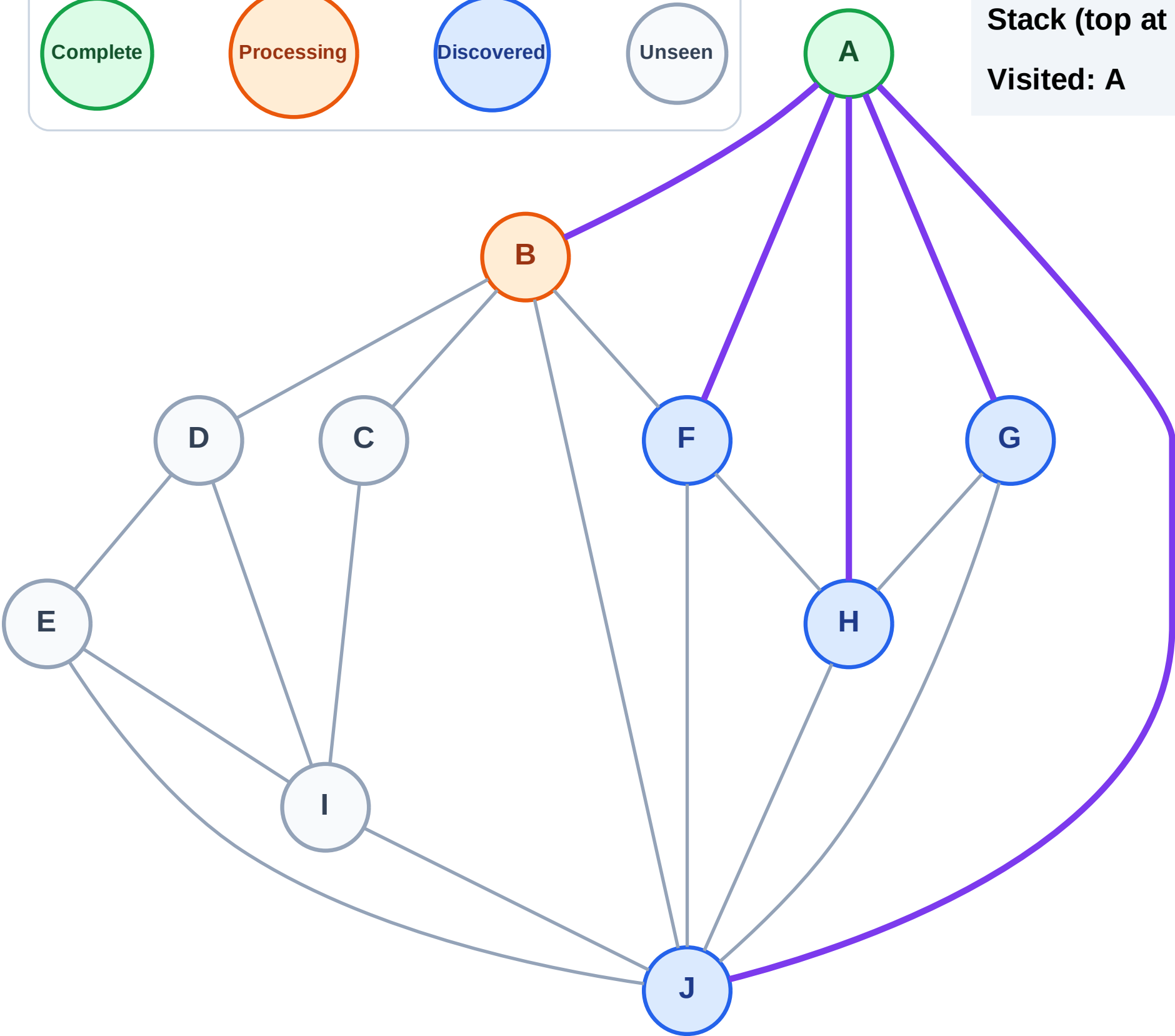
Step 4: Process node B

Legend



Stack (top at right): J | H | G | F

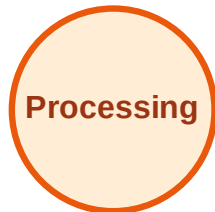
Visited: A



DFS Traversal

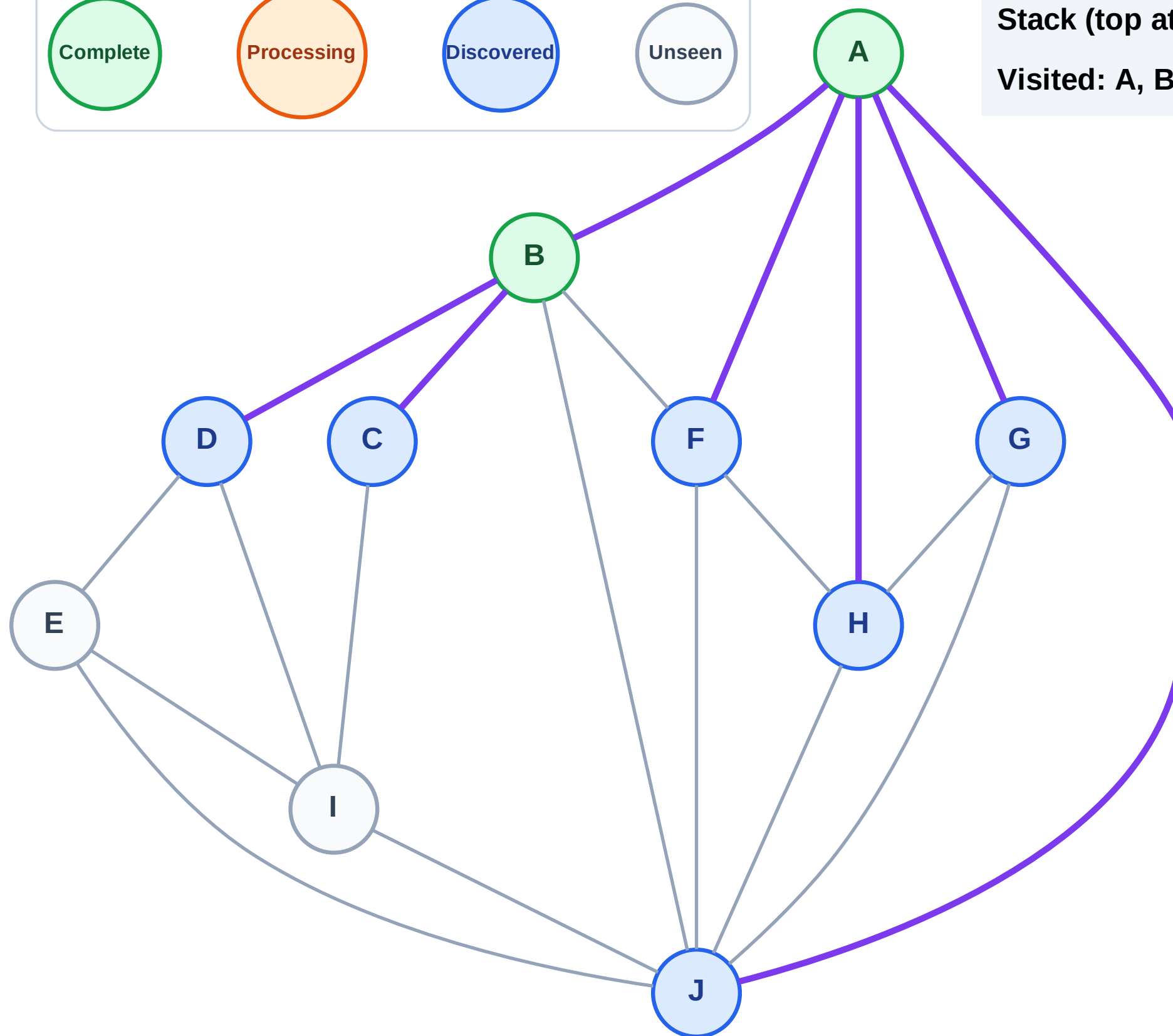
Step 5: Discover D, C from B

Legend



Stack (top at right): J | H | G | F | D | C

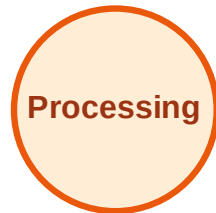
Visited: A, B



DFS Traversal

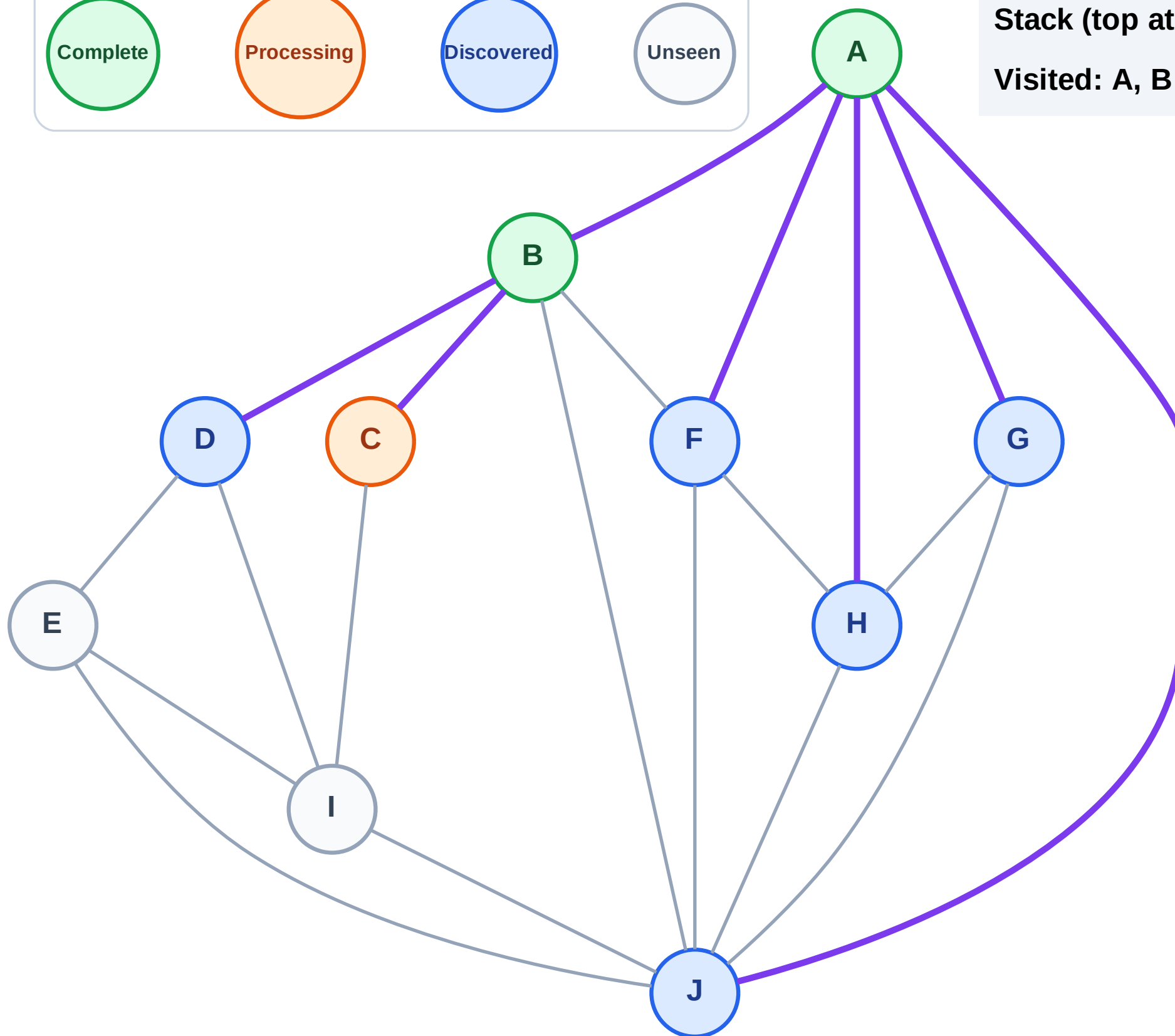
Step 6: Process node C

Legend



Stack (top at right): J | H | G | F | D

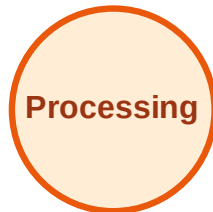
Visited: A, B



DFS Traversal

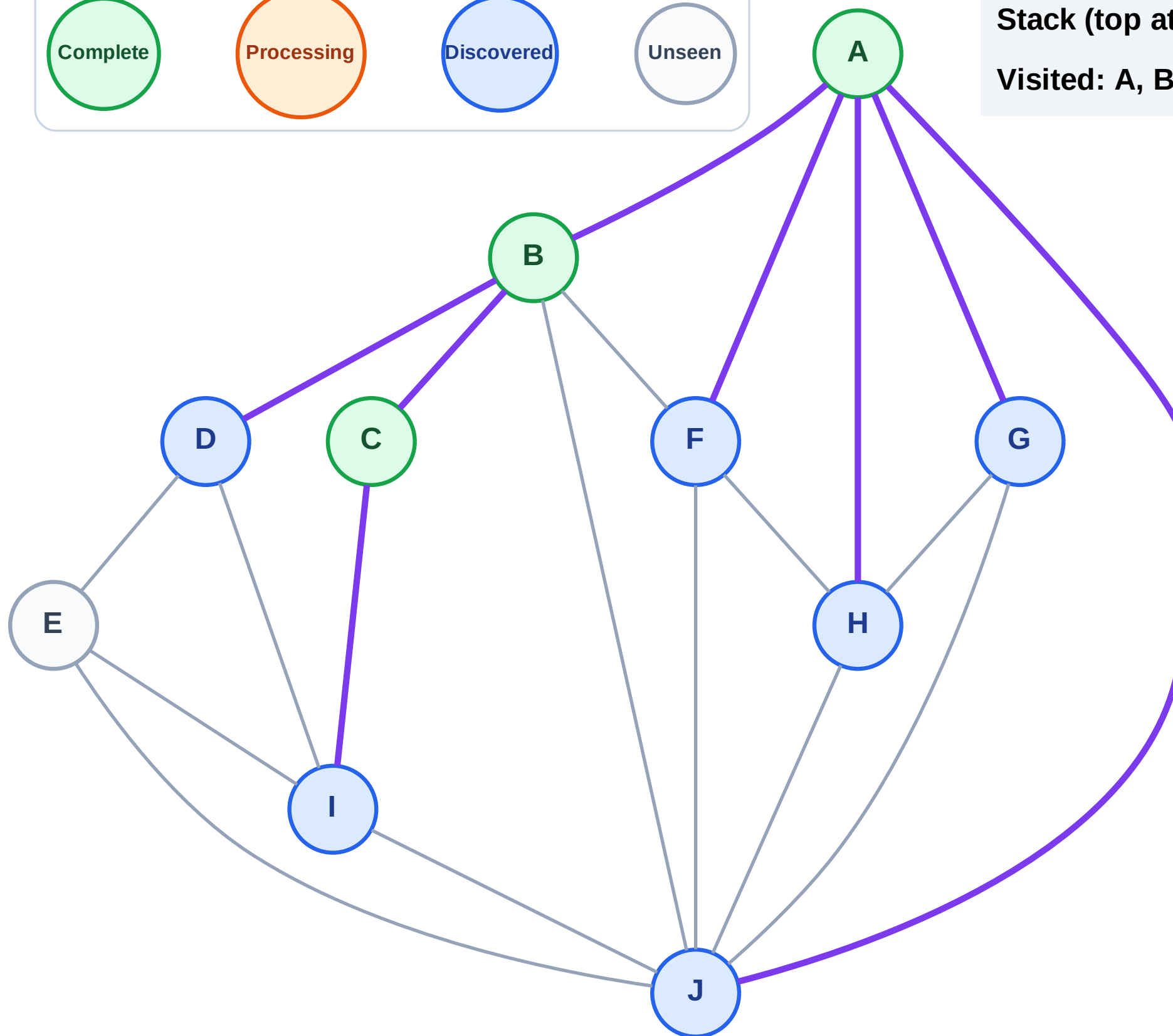
Step 7: Discover I from C

Legend



Stack (top at right): J | H | G | F | D | I

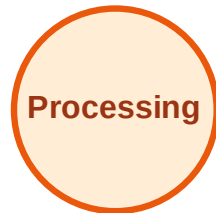
Visited: A, B, C



DFS Traversal

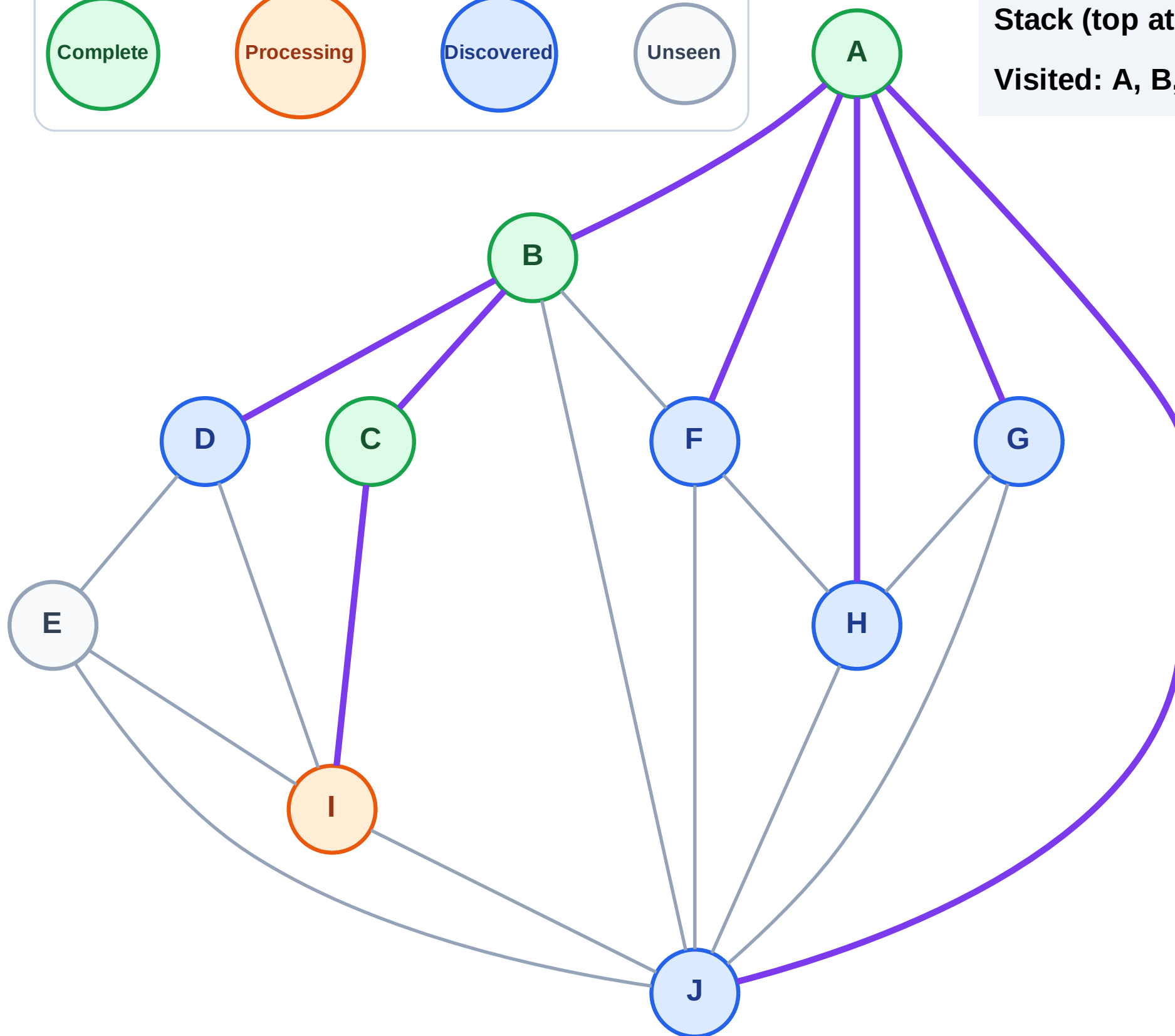
Step 8: Process node I

Legend



Stack (top at right): J | H | G | F | D

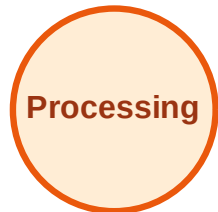
Visited: A, B, C



DFS Traversal

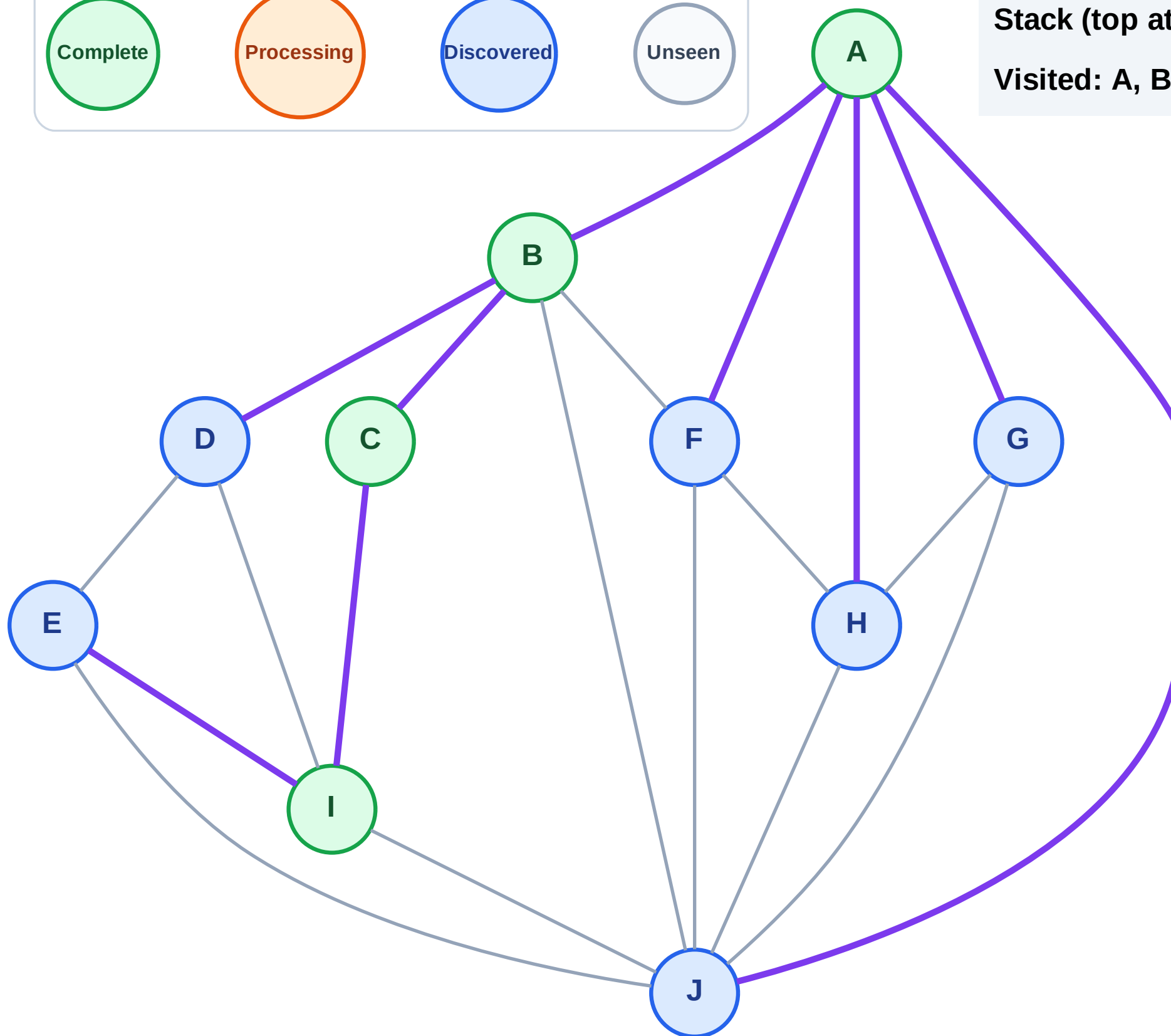
Step 9: Discover E from I

Legend



Stack (top at right): J | H | G | F | D | E

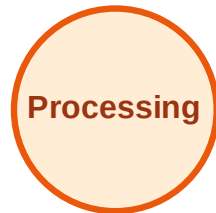
Visited: A, B, C, I



DFS Traversal

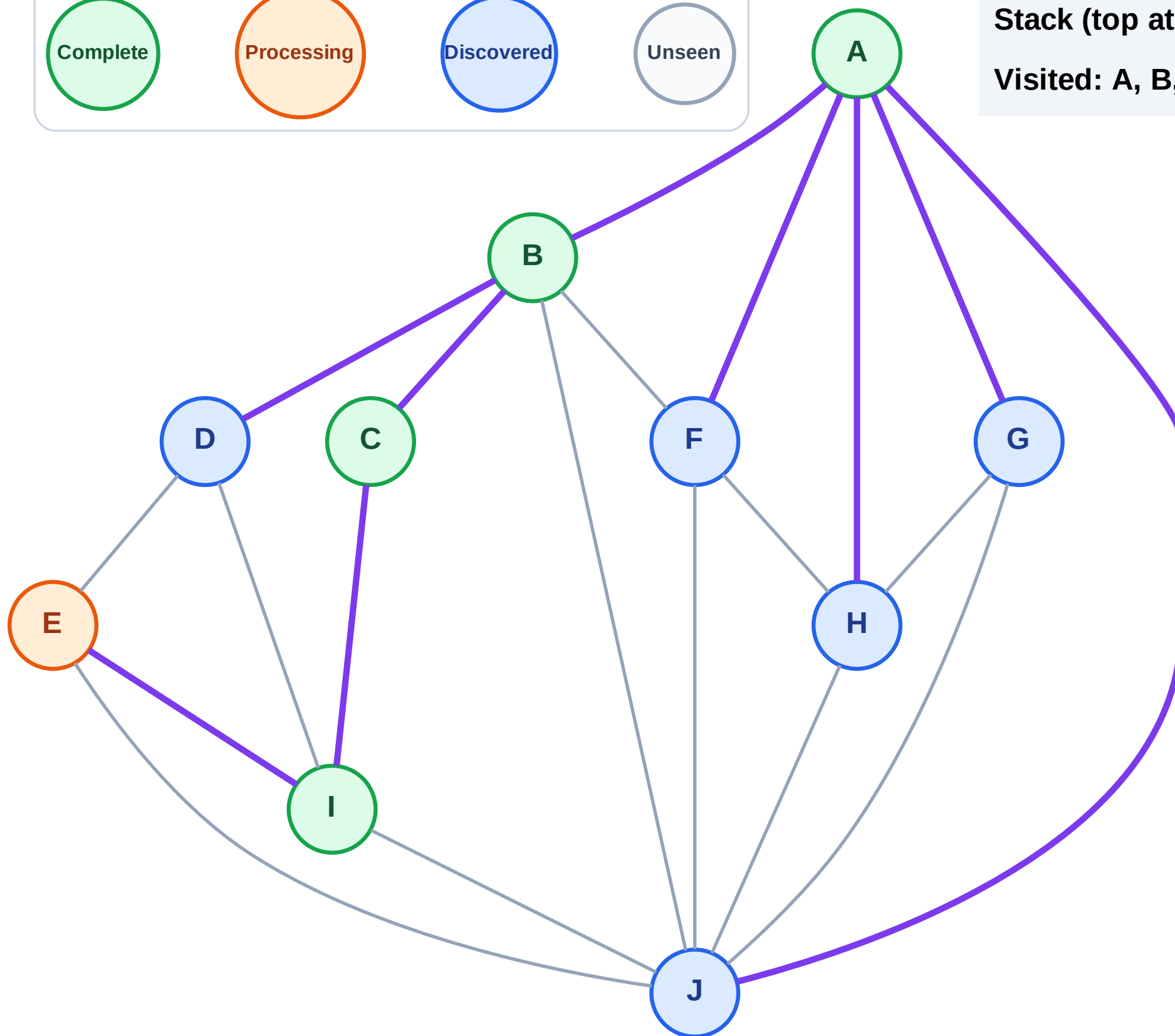
Step 10: Process node E

Legend



Stack (top at right): J | H | G | F | D

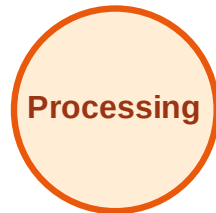
Visited: A, B, C, I



DFS Traversal

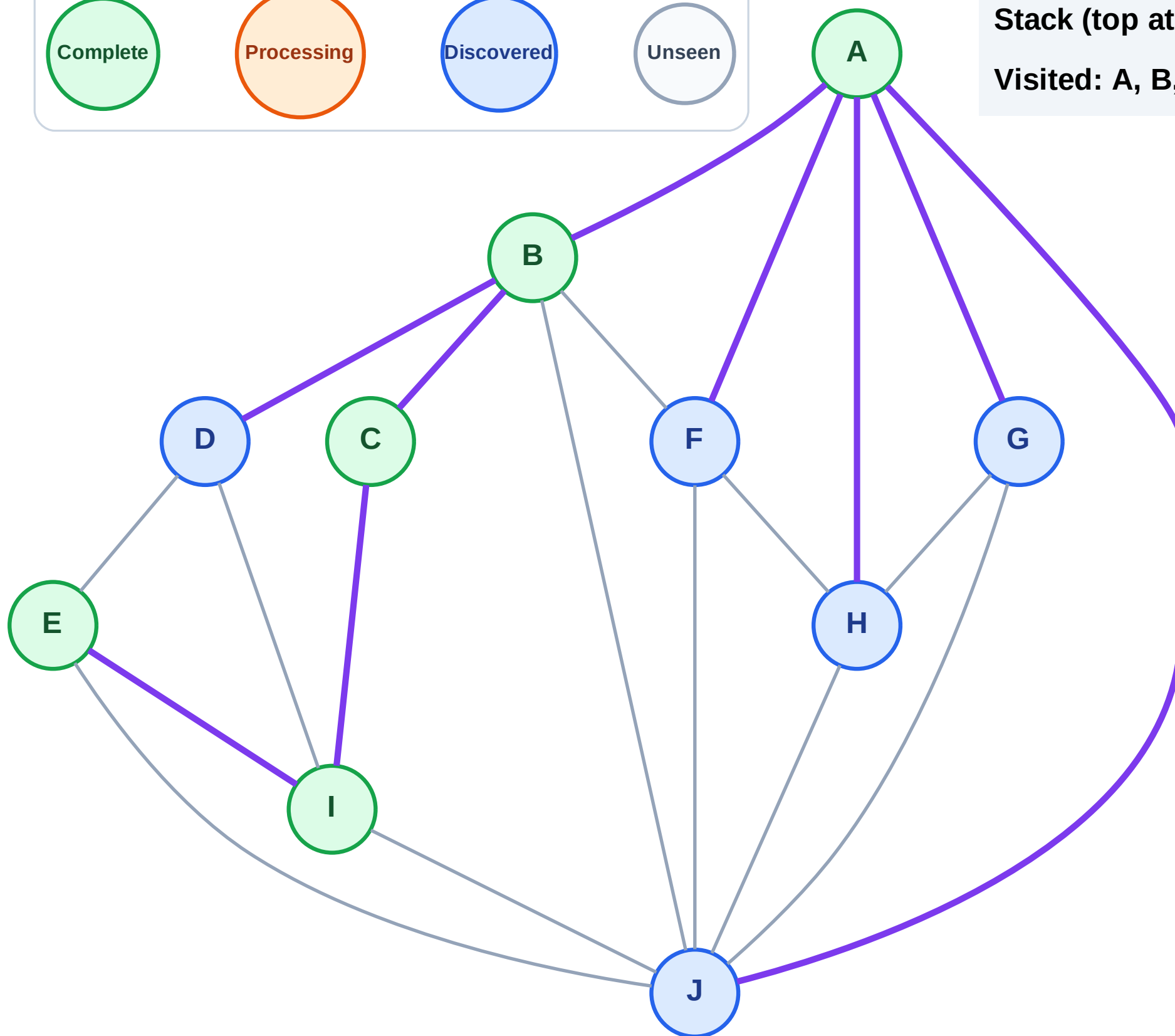
Step 11: No new neighbors from E

Legend



Stack (top at right): J | H | G | F | D

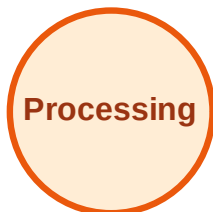
Visited: A, B, C, E, I



DFS Traversal

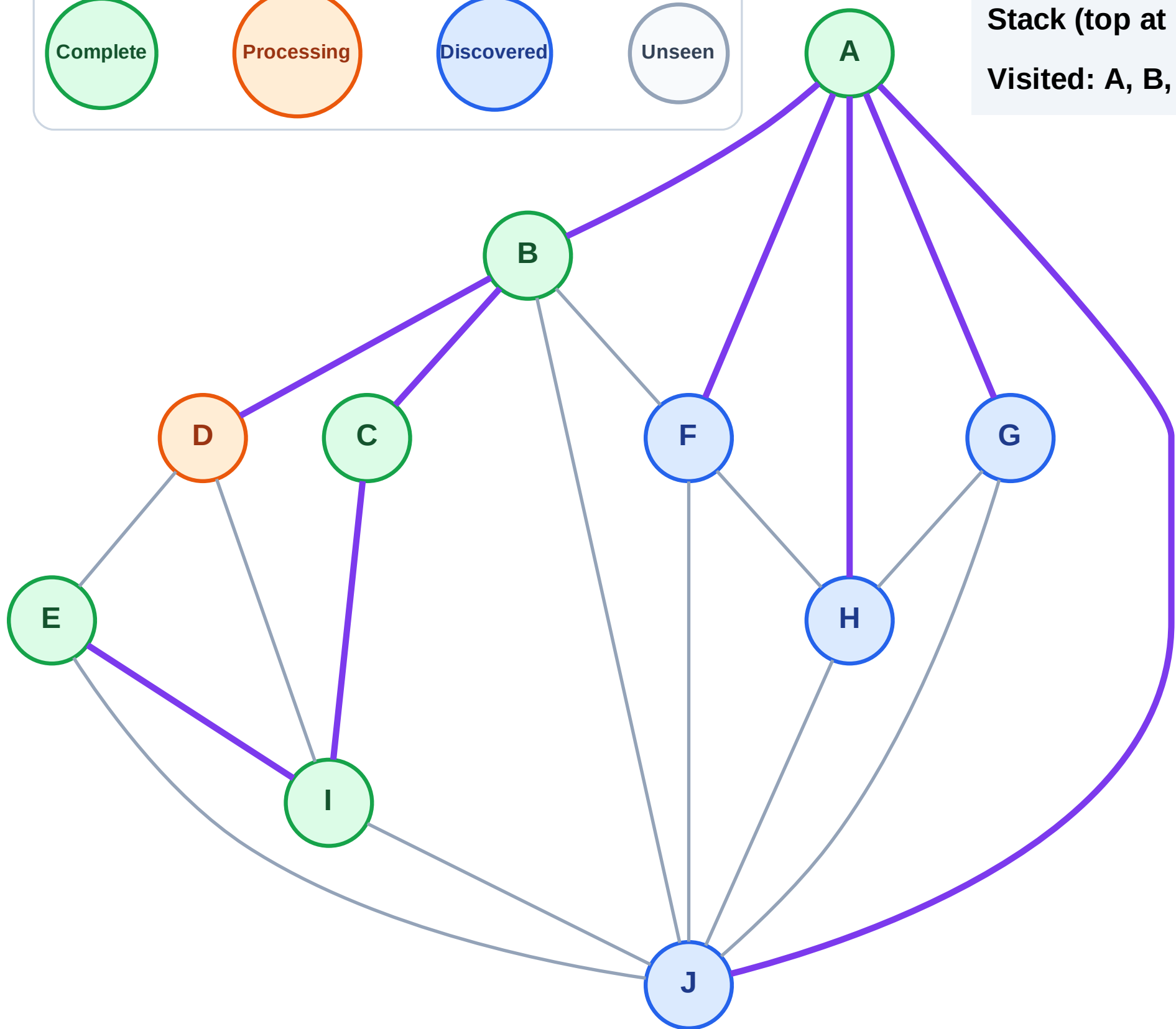
Step 12: Process node D

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, E, I



DFS Traversal

Step 13: No new neighbors from D

Legend

Complete

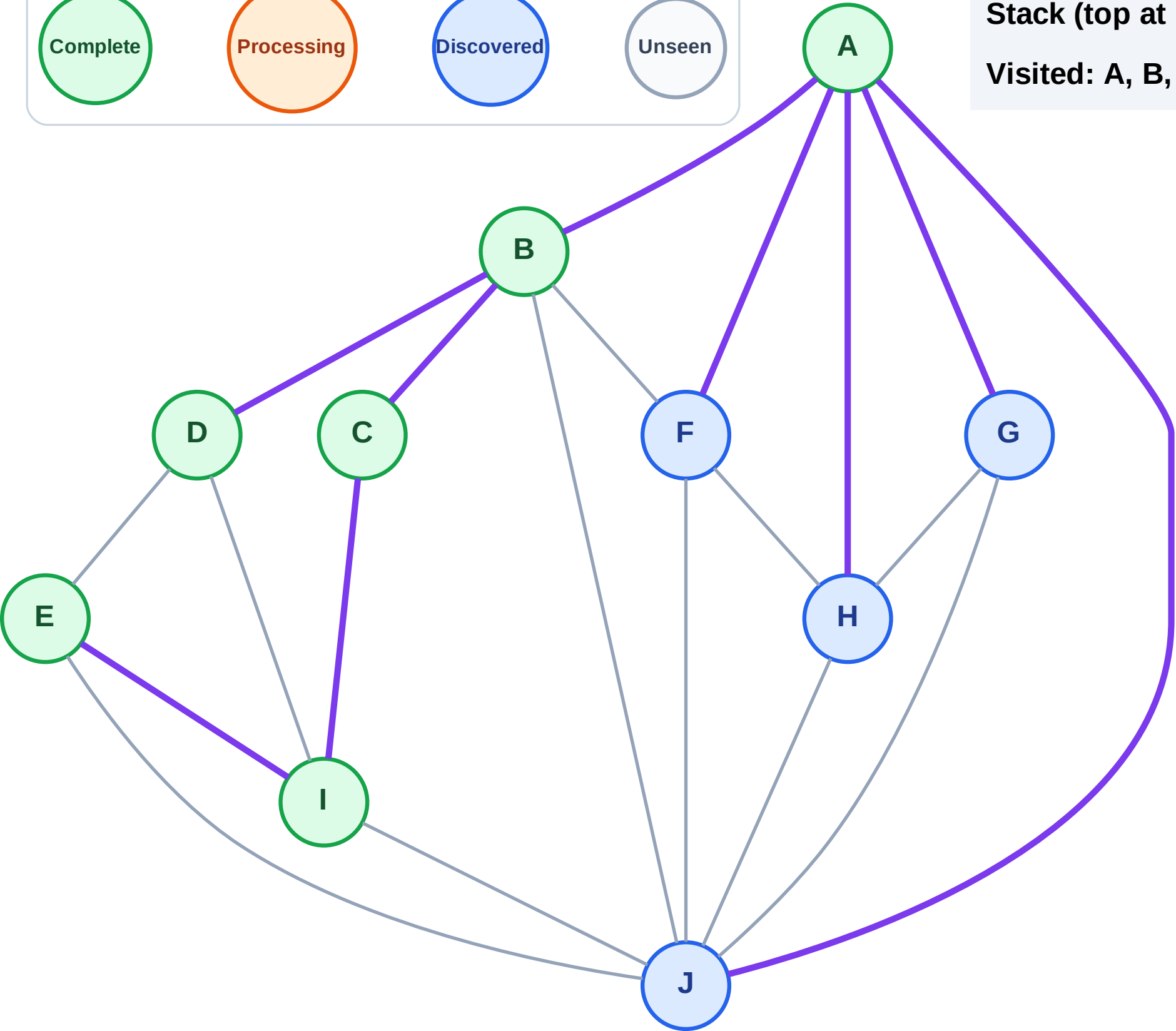
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F

Visited: A, B, C, D, E, I



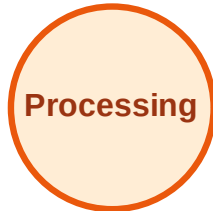
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



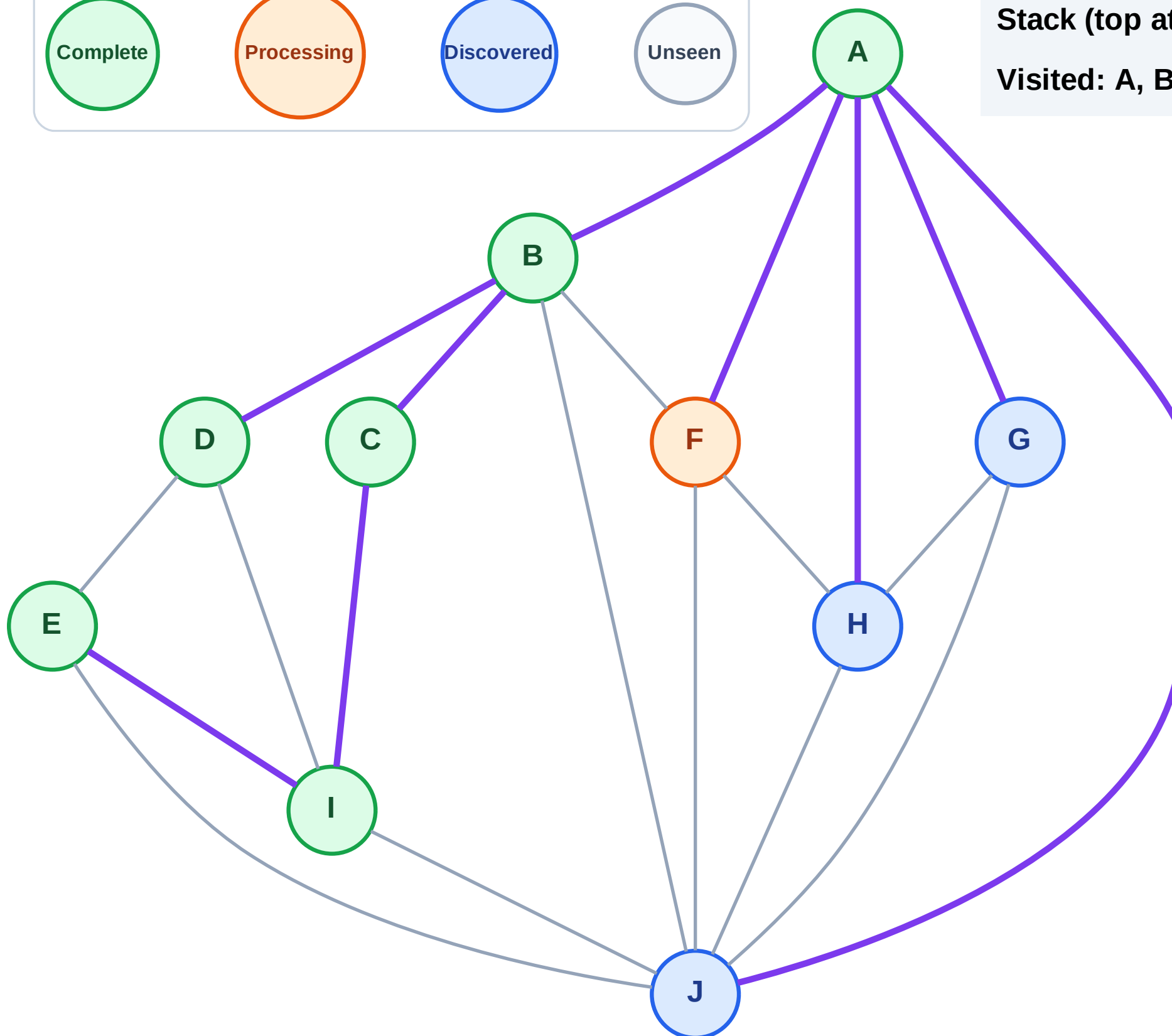
Discovered



Unseen

Stack (top at right): J | H | G

Visited: A, B, C, D, E, I



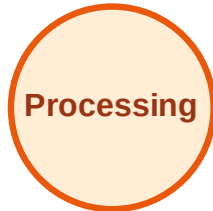
DFS Traversal

Step 15: No new neighbors from F

Legend



Complete



Processing



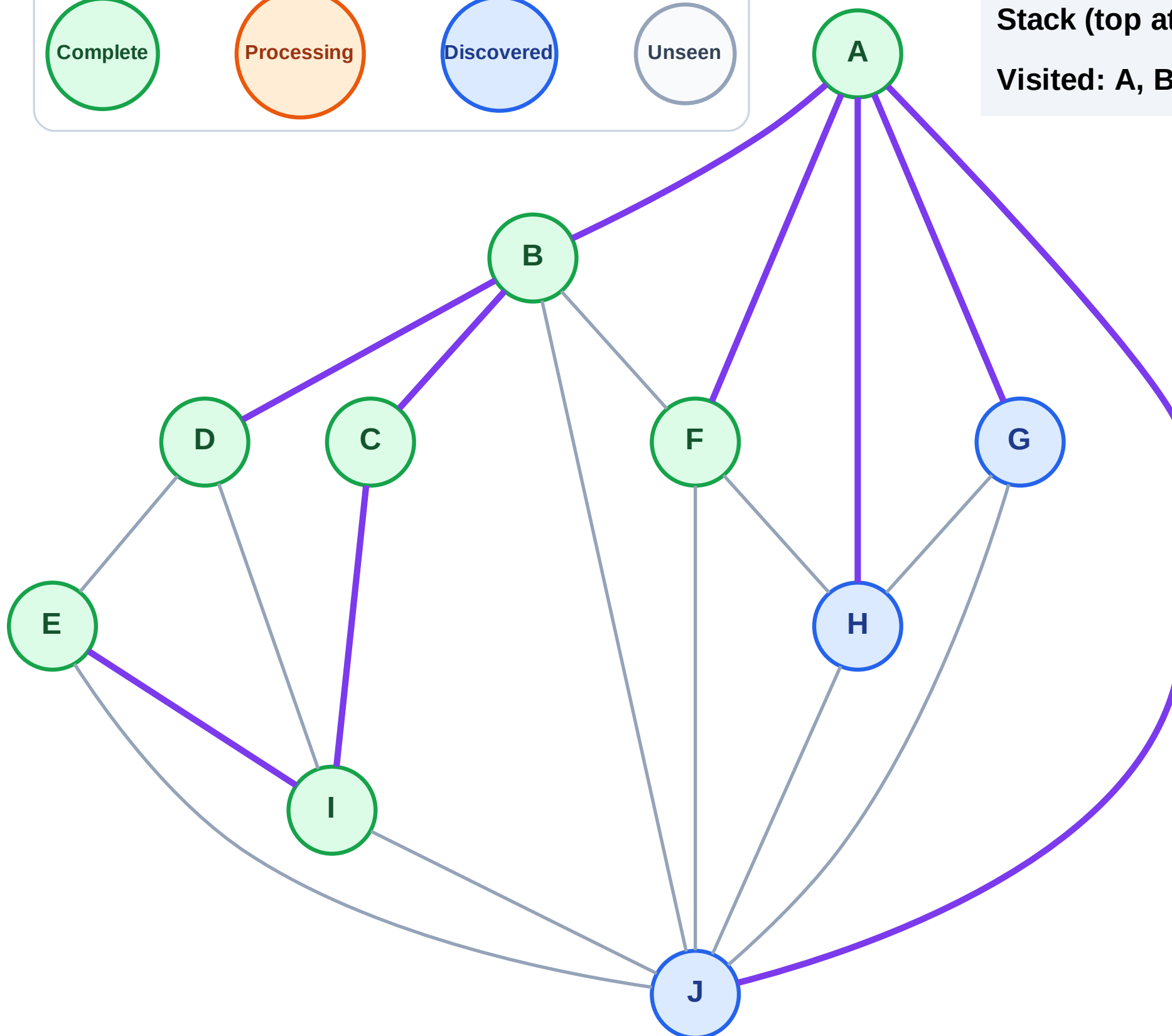
Discovered



Unseen

Stack (top at right): J | H | G

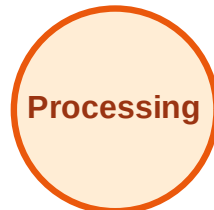
Visited: A, B, C, D, E, F, I



DFS Traversal

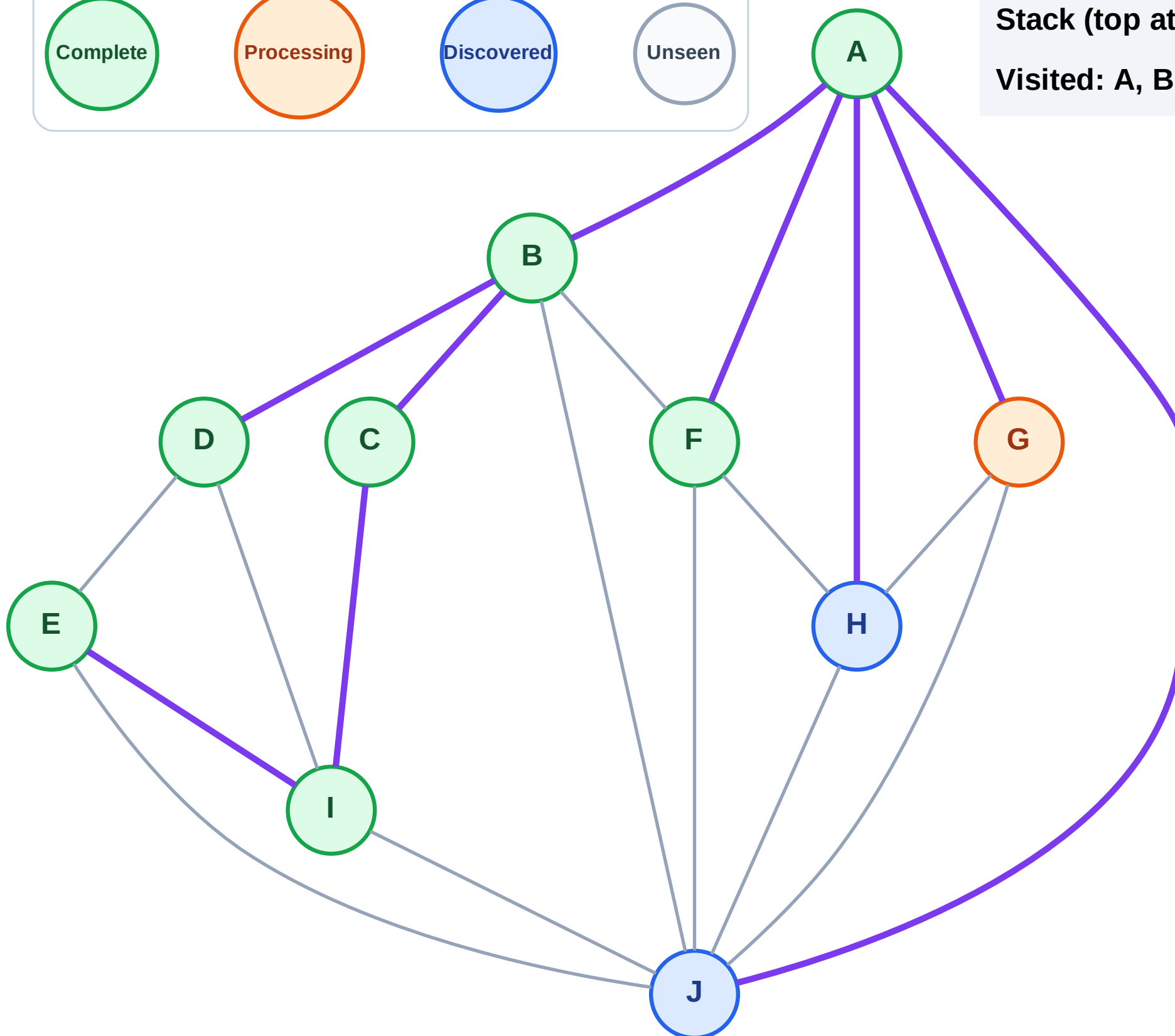
Step 16: Process node G

Legend



Stack (top at right): J | H

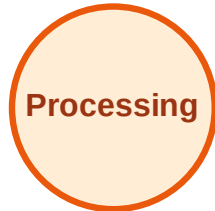
Visited: A, B, C, D, E, F, I



DFS Traversal

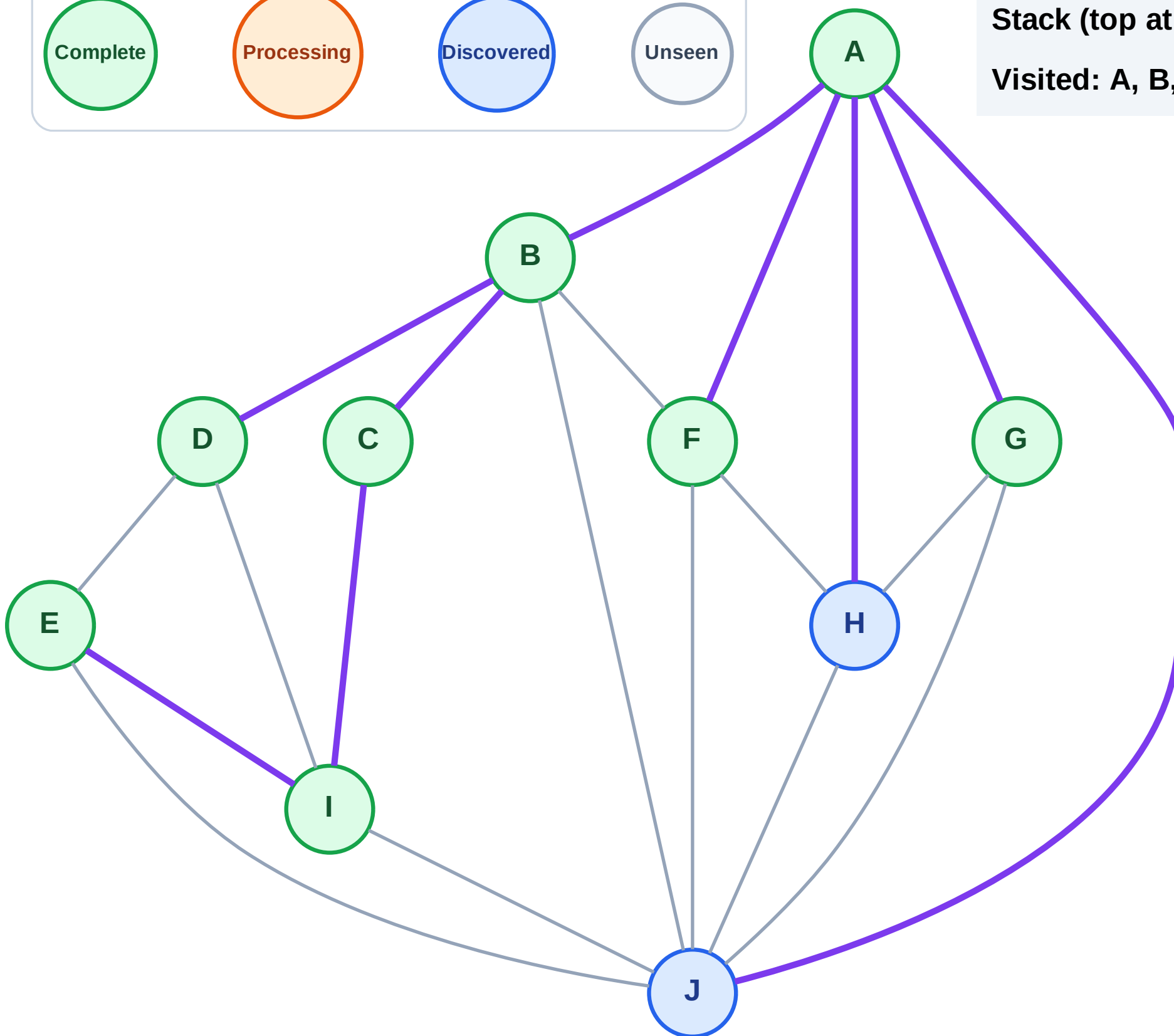
Step 17: No new neighbors from G

Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

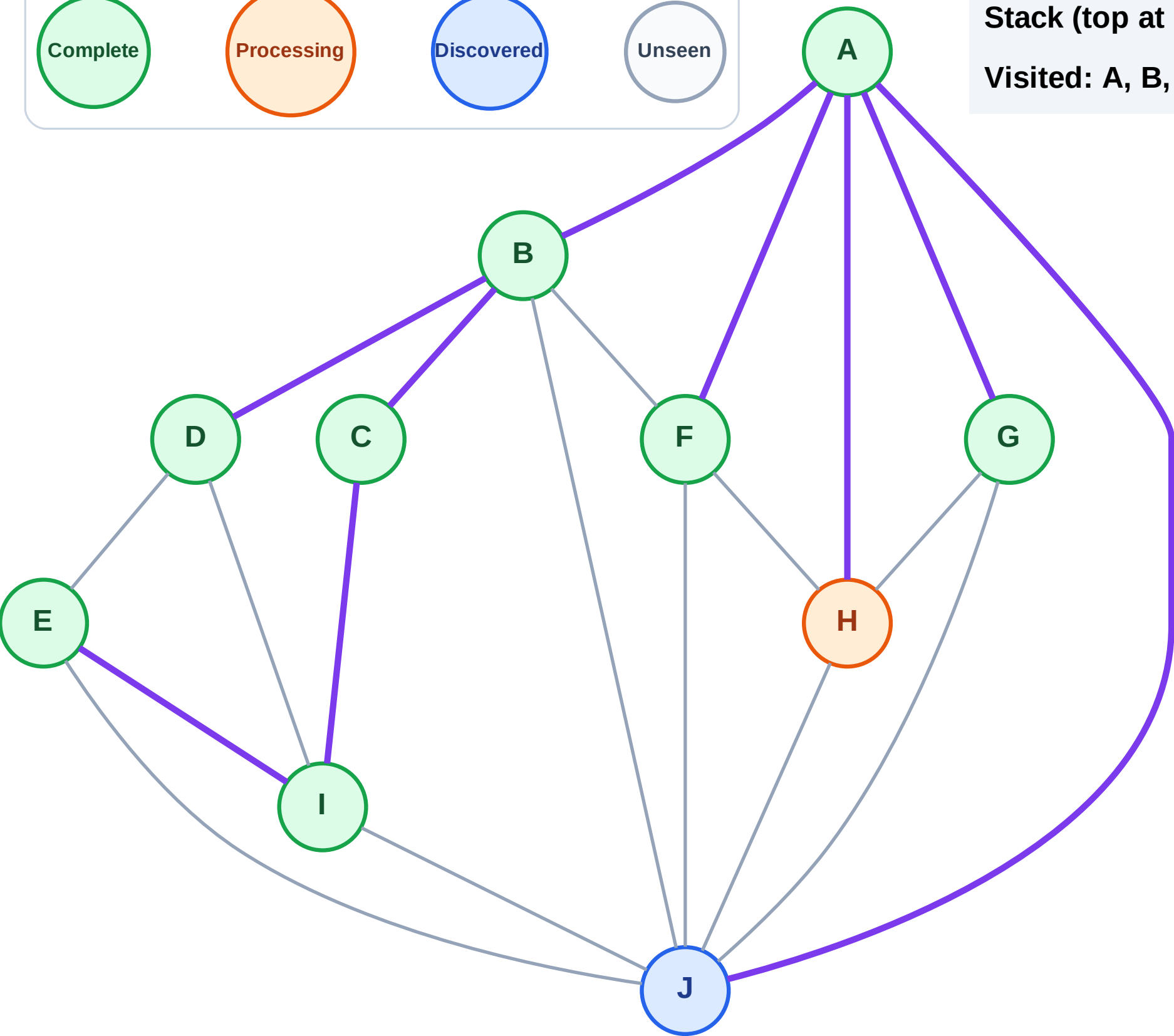
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, F, G, I



DFS Traversal

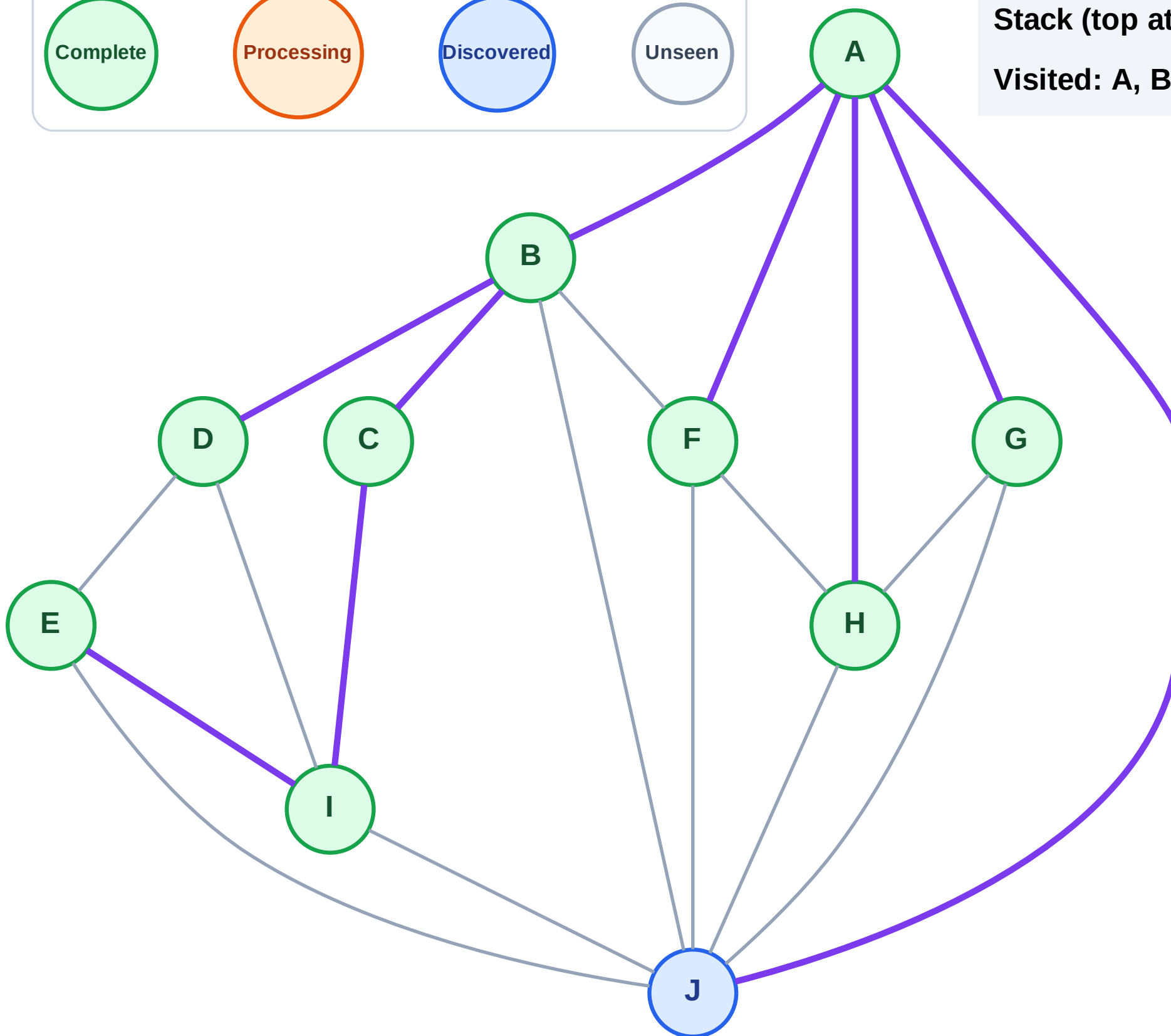
Step 19: No new neighbors from H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



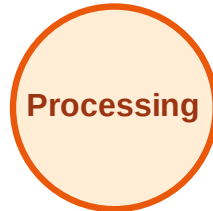
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



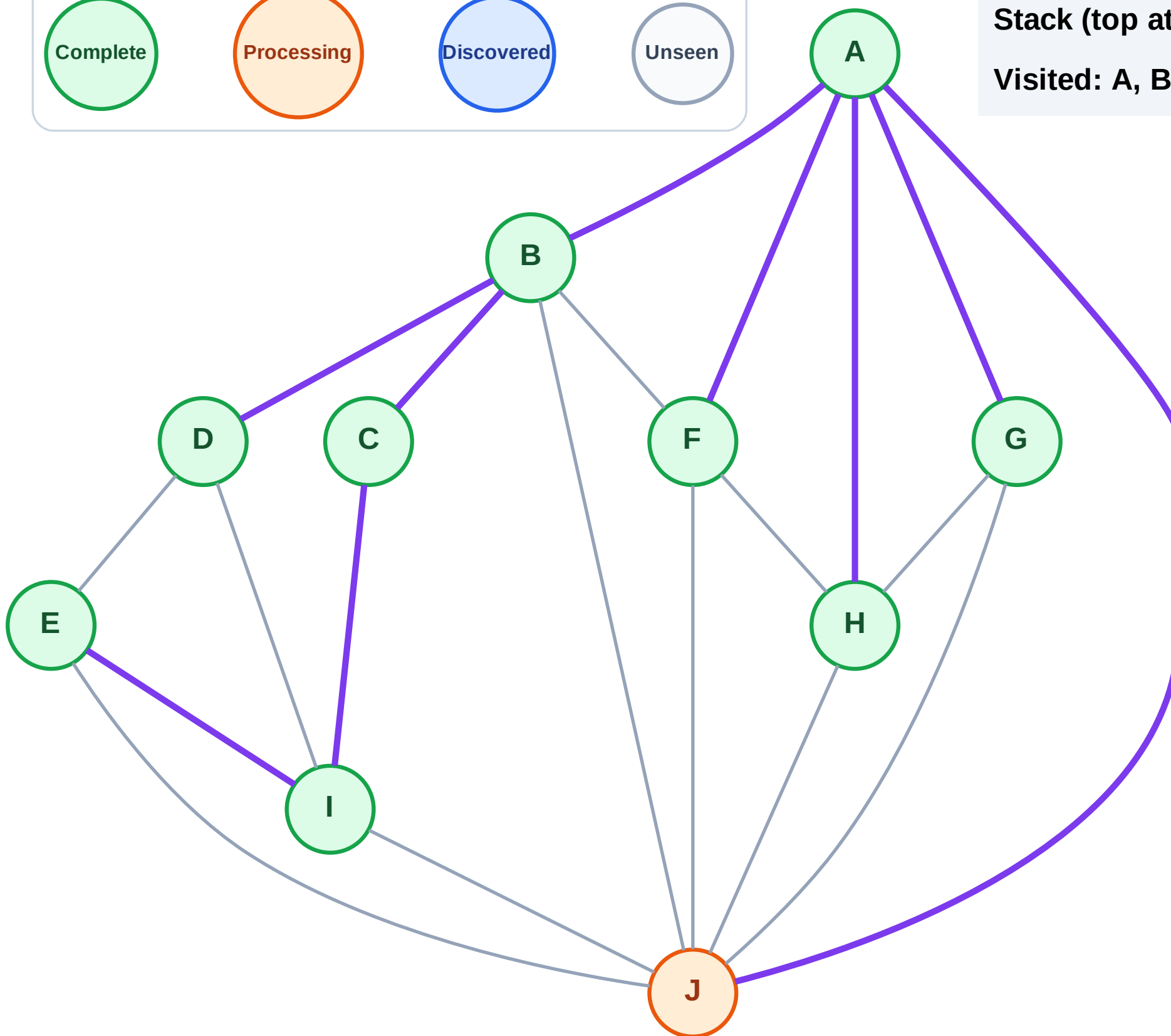
Discovered



Unseen

Stack (top at right): (empty)

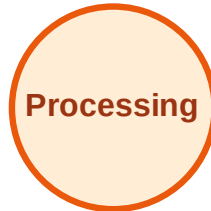
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

